

Linear Actuator LA36

Data Sheet



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Preface

Dear User,

We are delighted that you have chosen a LINAK® product.

LINAK systems are high-tech products based on many years of experience in the manufacture and development of actuators, lifting columns, desk frames, electric control boxes, controls, batteries, accessories and chargers.

This User Manual does not address the end user. It is intended as a source of information for the equipment or system manufacturer only, and it will tell you how to install, use and maintain your LINAK electronics. The manufacturer of the end product has the responsibility to provide a User Manual, where relevant safety information from this manual is passed on to the end user.

We are convinced that your LINAK product/system will give you many years of problem-free operation.

Before our products leave the factory, they undergo both function and quality testing. Should you, nevertheless, experience problems with your product/system, you are always welcome to contact your supplier.

LINAK subsidiaries and some distributors situated all over the world have authorised service centres, which are always ready to help you. Locate your local contact information on the back page.

LINAK provides a warranty on all products. (See warranty section).

This warranty, however, is subject to correct use in accordance with the specifications, maintenance being done correctly, and any repairs being carried out at a service centre, which is authorised to repair LINAK products.

Changes in installation and use of LINAK systems can affect their operation and durability. The products may only be opened by authorised personnel.

This User Manual has been written based on the present technical knowledge. LINAK reserves the right to carry out technical modifications and keeps the associated information updated.

LINAK A/S

Terms of use

LINAK® takes great care in providing accurate and up-to-date information on its products. However, the user is responsible for determining the suitability of LINAK products for a specific application.

Due to continual development, LINAK products are subject to frequent modifications and changes. LINAK reserves the rights to conduct modifications, updates, and changes without any prior notice. For the same reason, LINAK cannot guarantee the correctness and actual status of imprinted information on its products.

LINAK uses its best efforts to fulfil orders. However, for the reasons mentioned above, LINAK cannot guarantee availability of any particular product at any given time. LINAK reserves the right to discontinue the sale of any product displayed on its website or listed in its catalogues or in other written material created and produced by LINAK, LINAK subsidiaries, or LINAK affiliates.

All sales are subject to the 'Standard Terms of Sale and Delivery for LINAK A/S' available on LINAK websites.

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Introduction

Extremely powerful linear actuator made by LINAK. Up to 6,800 N thrust, or up to 160 mm/s. Designed to operate in extreme conditions. Solid choice for industrial and agricultural applications.

Safety instructions

Please read this safety information carefully.

Be aware of the following three symbols throughout the document:



Warning!

Failing to follow these instructions can cause accidents resulting in serious personal injury.



Recommendations

Failing to follow these instructions can result in the actuator suffering damage or being ruined.



Additional information

Usage tips or additional information that is important in connection with the use of the actuator.

Furthermore, ensure that all staff who are to connect, mount, or use the actuator are in possession of the necessary information and that they have access to this user manual.

Persons who do not have the necessary experience or knowledge of the product/products must not use the product/products. Besides, persons with reduced physical or mental abilities must not use the product/products, unless they are under surveillance or they have been thoroughly instructed in the use of the apparatus by a person who is responsible for the safety of these persons.

Moreover, children must be under surveillance to ensure that they do not play with the product.

Before you start mounting/dismounting, ensure that the following points are observed:

- The actuator is not in operation.
- The actuator is free from loads that could be released during this work.

Before you put the actuator into operation, check the following:

- The actuator is correctly mounted as indicated in the relevant user instructions.
- The equipment can be freely moved over the actuator's whole working area.
- The actuator is connected to a mains electricity supply/transformer with the correct voltage and which is dimensioned and adapted to the actuator in question.
- Ensure that the voltage applied matches to the voltage specified on the actuator label.
- Ensure that the connection bolts can withstand the wear.
- Ensure that the connection bolts are secured safely.

During operation, please be aware of the following:

- Listen for unusual sounds and watch out for uneven running. Stop the actuator immediately if anything unusual is observed.
- Only use the actuator within the specified working limits.
- Do not step on or kick the actuator.

When the equipment is not in use:

- Switch off the mains supply in order to prevent unintentional operation.
- Check regularly for extraordinary wear.

Classification

The equipment is not suitable for use in the presence of a flammable anaesthetic mixture with air or with oxygen or nitrous oxide.

**Warnings**

- Do not sideload the actuator.
- When mounting the actuator in the application ensure that the bolts can withstand the wear and that they are secured safely.
- If irregularities are observed, the actuator must be replaced.
- The standard actuator (without Integrated Controller) without clutch, is not allowed to run into a mechanical block -before reaching the end of stroke.

**Recommendations**

- Do not place load on the actuator housing.
- Prevent impact or blows, or any other form of stress to the housing.
- Ensure that the cable cover is mounted correctly. Use 3.5 Nm torque.
- Ensure that the duty cycle and the usage temperatures for LA36 actuators are respected.
- Ensure that the cable cannot be squeezed, pulled or subjected to any other stress.
- Furthermore, it will be good practice to ensure that the actuator is fully retracted in the “normal” position. The reason is that there will be a negative pressure inside the actuator if it is extended which over time can lead to water entering the actuator.

Features

- 12, 24, 36 or 48 V DC permanent magnetic motor (IC only 12/24 V DC)
- Load from 500 N - 6.800 (depending on gear ratio and spindle pitch)
- Max. speed 160 mm/sec. (depending on gear ratio and spindle pitch)
- Stroke length from 100 to 1200 mm
- Built-in endstops reached function
- Highly efficient acme thread spindle
- Safety factor 2: The actuator has been certified to withstand static loads that are twice the magnitude of its dynamic load capacity without sustaining damage.
- Heavy duty aluminium housing for harsh conditions
- Protection class: IP66 for outdoor use (dynamic). Furthermore, the actuator can be washed down by a high pressure cleaner (IP69K - static)
- Hand crank for manual operation
- Integrated brake with high self-lock ability
- Endplay: see [Technical Specifications](#)
- Non-rotating piston rod eye
- Noise level: 76 dB (A). Measuring method: DS/EN ISO 3746 (actuator not loaded)
- Off-highway Features:
 - 12 or 24 V DC brushed permanent magnetic motor
 - Load from 1,700 N - 6,800 N (depending on gear ratio and spindle pitch)
 - Max. speed 7 mm/sec. to 100 mm/sec. (depending on gear ratio and spindle pitch)
 - Reinforced aluminium housing for harsh conditions
 - IPC-A-610 Class 3 (High-performance electronic products)
 - IP54 without cable mounted
IP69K (static) with cable mounted with shell or moulded cable

An Off-highway vehicle is intended for use on steep or uneven ground and includes those used for construction or agriculture. They are specifically designed for off-road use.

Quad bikes, dirt bikes, dune buggies and other types of all-terrain vehicles are also types of Off-highway vehicles, although their function is very different from motor vehicles designed for industrial and farming use.

Options in general

- Back fixture can be ordered in steps of 30 degrees
- Exchangeable cables in different lengths
- Hall effect sensor
- Analogue or digital feedback for precise positioning
- Endstop reached signals
- Mechanical overload protection through integrated slip clutch -Standard actuators only
- Mechanical potentiometer (not applicable with IC)
- When ordering AISI (304 and up) piston rod eye and back fixture, stainless steel screws are automatically included
- Special anodised housing for extreme environments -see paragraph regarding 'Special anodised housing'
- Adjustable magnetic sensors for endstop signals (code no. 1017031)
- IC options including:
 - I/O
 - Ethernet/IP
 - Modbus TCP/IP
 - Modbus RTU
 - PROFINET
 - IO-Link
 - LIN bus
 - CAN SAE J1939
 - CANopen
 - Off-highway LIN bus
 - Off-highway CAN SAE J1939
 - Off-highway CANopen

See specific interface user manuals at the [TECHLINE webpage](#) for Connection Diagrams and I/O specifications

- PC configuration tool (BusLink or Actuator Connect™)
- ATEX/IECEx/CCC (Ex) certified for Zone 21 (Not applicable for Off-highway actuators)

Usage

- Duty cycle up to 600 mm stroke: max. 20% (4 min. drive and 16 min. rest)
 - Duty cycle at 601-999 mm stroke: max. 15% (3 min. drive and 17 min. rest)
 - Duty cycle at 1000-1200 mm stroke: max. 10% (2 min. drive and 18 min. rest)
 - Ambient operating temperature: Full performance from +5°C to +40°C
-30°C (reduced load 50%) to + 85°C (reduced duty cycle 10%)
-40°C (no load)
 - Ambient operating temperature for ATEX/IECEX/CCC: -25°C to +65°C
 - Storage temperature: -40 °C to +70 °C
Actuator is not activated/
connected -40°C to +85°C for 72 hours
-55°C to +95°C for 24 hours for Standard platform
-55°C to +105°C for 24 hours for Integrated Control platform
Acclimatization before usage.
 - Relative humidity: Full performance from 20 % to 80 % - non-condensing
(Actuator is neither activated nor connected)
 - Cyclic state: 93 % to 98 % - non-condensing +25°C to +55°C for 12 hours
 - Steady state: 93 % to 95 % - non-condensing +40°C for 56 days
 - Atmospheric pressure: 700 to 1060 hPa
 - Meters above sea level: Max. 3000 meters
- For applications operated at constantly low temperatures it might be beneficial to recommend a stronger version of the actuator to reduce the current consumption that in some combinations can be up to 3 times higher (at -40° Celsius)
- Off-highway:
 - Testet according to: ISO14982-1 / Agricultural and forestry machinery - Electromagnetic compatibility - Part 1: General EMC requirements (clamped capacitor circuit)
 - Compliant with: ROHS2 : 2011/65/EU: Directive on the restriction of the use of certain hazardous substances in electrical and electronic equipment EMC Directive - 2014/30/EU

Ordering Example

36 120 200 0 A 01 B 6 - 6 1 1 H 3 0300 N C S 0 0 0

Actuator type	36	= LA36		
Spindle type	080	= 8 mm	120	= 12 mm
	160	= 16 mm	200	= 20 mm
Stroke length	200	= XXX Length in mm (50-995)	AXX	= 10XX Length in mm (1,000-1,095)
	BXX	= 11XX Length in mm (1,100-1,195)	C00	= 1,200 mm
Safety	0	= No safety nut	A	= Safety nut (Push only)
Feedback	0	= No Feedback	K	= Single Hall
	A	= Analogue feedback (Hall potentiometer)	I	= Potential Free
	F	= PWM	P	= Potentiometer (standard platform actuators only)
	H	= Dual Hall	X	= Special
Platform	6-pin		9-pin	
	Endstop switch principle		Zero Point	
	00	= Standard	B3	= I/O Basic
	01	= Standard with power switch	C3	= I/O Customised
	07	= CAN SAE J1939	F3	= I/O Full
	08	= CANopen	0B	= IO-Link
			14	= Modbus RTU
	Zero Point		Zero Point with split power supply	
	16	= LIN bus	A7	= CAN SAE J1939
	17	= CAN SAE J1939	A8	= CANopen
	18	= CANopen	0E	= Modbus TCP/IP*
			2E	= EtherNet/IP*
			4E	= PROFINET*

* Not available with ATEX/IECEX/CCC approval

18-pin Off-highway**

C6**	= LIN bus*
D6**	= CAN SAE J1939*
E6**	= CANopen*
XX	= Special*

** Requires Housing option 'C' IP66 Off-highway, also only available with Motor Type 1 or 2

Motor type	A	= 12 V DC with Clutch	1	= 12 V DC
	B	= 24 V DC with Clutch	2	= 24 V DC
	C	= 36 V DC with Clutch	3	= 36 V DC
	J	= 48 V DC with Clutch*	4	= 48 V DC*

Housing	6	= IP66 - Reinforced house	A	= IP66
	9*	= Harsh environment	T**	= IP66 ATEX / IECEx / CCC approved
			C***	= IP66 Off-highway
Reed	-	= Without Reed limit switch	+	= With Reed limit switch
Colour	6*	= Dark Olivish Grey NCS S7000-N	X	= Special
Back fixture	1	= 0 °	A	= 30°
	2	= 90°	B	= 60°
	4	= Male Adapter (Outer thread)	C	= 120°
	5	= Female Adapter (Inner thread)	D	= 150°
	6	= Rotated in 30° intervals	X	= Special
Piston rod eye	1	= Slotted	5	= Female Adapter (Inner thread)
	2	= Solid	6	= Ball eye
	4	= Male Adapter (Outer thread)	X	= Special
Gear	E	= Ratio 1:7	F	= Ratio 1:18
	G	= Ratio 1:31	H	= Ratio 1:46
Brake	3	= Push/Pull		
Built-in dimension	0300	= 300 mm (min. length)	xxxx	= Measured in mm
Endstop reached output In/Out	A	= A_HIGH / A_HIGH	J	= A_HIGH / LOW
	B	= A_LOW / A_HIGH	K	= A_LOW / LOW
	C	= A_HIGH / A_LOW	L	= A_HIGH / HIGH
	D	= A_LOW / A_LOW	M	= A_LOW / HIGH
	E	= LOW / A_HIGH	N	= LOW / LOW
	F	= HIGH / A_HIGH	O	= HIGH / LOW
	G	= LOW / A_LOW	P	= LOW / HIGH
	H	= HIGH / A_LOW	Q	= HIGH / HIGH
			X	= Special
Plug type	0*	= No plug (requires cable '0')	H	= AMP
	J	= Deutsch	K	= AMP Super Seal
	9	= Deutsch - Moulded	7	= AMP Super Seal - Moulded
	C	= Flying leads**	E***	= M12 Y Ethernet
	N	= M12 IO-Link	R	= M12 Modbus RTU
			X	= Special

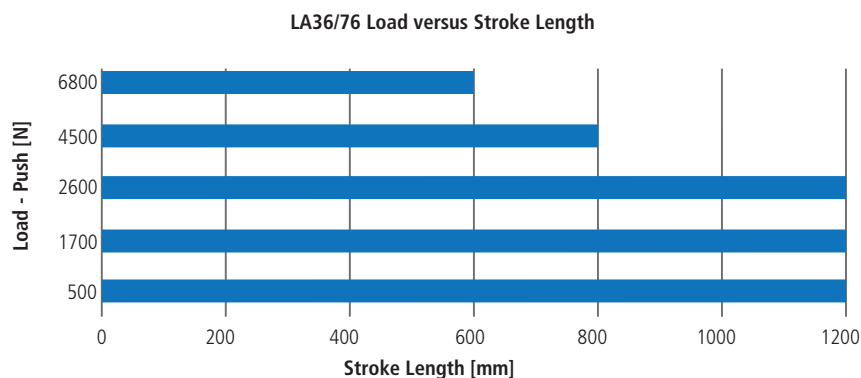
Cable	0*	= No cable selected	A	= Mounted with 90° angled connectors
	S***	= Straight cable**	Y	= Y-Cable (combined power and signal cable)
			X	= Special
Parallel mode	0	= Parallel disabled	2-8	= Critical parallel (number of actuators in the parallel system)
SW config.	0	= Default software	X	= Special software
Not used	0	= Not used		

* Shall be chosen with platform: '18-pin Off-highway'

** Required for ATEX/IECEX/CCC approval

*** Actuators with Ethernet/IP are only compatible with Plug type E and Cable type S (ordering number: CAB0367402)

Load vs stroke length



Please note:

500-1,700 N is with 20 mm spindle pitch

500-5,600 N is with 16 mm spindle pitch

500-6,800 N is with 12 mm spindle pitch

500-6,800 N is with 8 mm spindle pitch

- For applications that only operate in pull, the limitations are 1,200 mm stroke and 6,800 N load

Technical specifications

12 V

Load max. (N)	Self-lock min. (N)	Pitch (mm/ spindle rev.)	Gear/ Ratio	Hall Resolution mm/count	End-play in mm	*speed (mm/s)			Standard stroke lengths (mm) in steps of 50 mm	*Consumption (A)	
						@ 300 N	No-min-al load	Reg-ula-ted		@ 300 N	No-min-al load
500***	1,000	20	E 1:7	0.721	4.5	145	135	114	100-1,200	4.5	20
1,700	2,200	20	F 1:18	0.721	3.5	67	52	52	100-1,200	7.9	23
2,000	2,600	16	F 1:18	0.577	2.9	54	43	44	100-1,200	6.0	23
2,600	3,400	12	F 1:18	0.433	2.6	40	34	34	100-1,200	5.6	24
3,400	4,400	16	G 1:31	0.339	2.3	30.8	25	25	100-1,200**	5.5	23
4,500	5,800	12	G 1:31	0.254	2.3	24.3	18.5	18	100-1,200**	5.6	24.5
5,600	6,600	16	H 1:46	0.221	2.2	20.7	16	16	100-1,200**	5.5	24
6,800	8,800	12	H 1:46	0.166	2.2	16	13.2	13	100-1,200**	5.0	23.5
6,800	13,000	8	H 1:46	0.110	2.2	11	9	8.8	100-1,200**	4.7	17.0

24 V

Load max. (N)	Self-lock min. (N)	Pitch (mm/ spindle rev.)	Gear/ Ratio	Hall Resolution mm/count	End-play in mm	*speed (mm/s)			Standard stroke lengths (mm) in steps of 50 mm	*Consumption (A)	
						@ 300 N	No-min-al load	Reg-ula-ted		@ 300 N	No-min-al load
500***	1,000	20	E 1:7	0.721	4.5	145	135	114	100-1,200	2.4	10.0
1,700	2,200	20	F 1:18	0.721	3.5	68	52	54.4	100-1,200	2.4	10.3
2,000	2,600	16	F 1:18	0.577	2.9	54.7	43	43.7	100-1,200	2.4	10.3
2,600	3,400	12	F 1:18	0.433	2.6	41	33	32.8	100-1,200	2.4	10.4
3,400	4,400	16	G 1:31	0.339	2.3	31.1	25	24.8	100-1,200**	2.4	10.3
4,500	5,800	12	G 1:31	0.254	2.3	23.3	18.9	18.6	100-1,200**	2.4	10.2
5,600	6,600	16	H 1:46	0.221	2.2	21	17	16.8	100-1,200**	2.4	10.3
6,800	8,800	12	H 1:46	0.166	2.2	15.7	12.7	12.5	100-1,200**	2.4	10.3
6,800	13,000	8	H 1:46	0.110	2.2	11	7	8.8	100-1,200**	2.4	8

* The typical values can have a variation of $\pm 20\%$ on the current values and $\pm 10\%$ on the speed values. Measurements are made with an actuator in connection with a stable power supply and an ambient temperature at 20°C.

** There are limitations on the stroke length. If you need full load, please see: "Load v. Stroke Length".

*** Note: Fully loaded actuators need a soft start in order to prevent the clutch from slipping when starting (see curves). (Not optional for 'Off-highway' actuators)



Please note that all actuators featuring 'LIN bus', 'CAN SAE J1939', 'CANopen' and 'Modbus' will run at a regulated speed, see table above.

Technical specifications:

36 V

Load max. (N)	Self-lock min. (N)	Pitch (mm/ spindle rev.)	Gear/ Ratio	Hall Resolution mm/count	End-play in mm	*Typical speed (mm/s)			Standard stroke lengths (mm) in steps of 50 mm	*Typical amp. (A)	
						No load	Full load	Reg- ula- ted		No load	Full load
500***	1,000	20	E 1:7	0.721	4.5	160	135		100-1,200	2.0	8.0
1,700	2,200	20	F 1:18	0.721	3.5	68	52		100-1,200	2.0	8.0
2,000	2,600	16	F 1:18	0.577	2.9	54.7	43		100-1,200	2.0	8.0
2,600	3,400	12	F 1:18	0.433	2.6	41	33.5		100-1,200	2.0	8.0
3,400	4,400	16	G 1:31	0.339	2.4	31.1	25		100-1,200**	2.0	8.0
4,500	5,800	12	G 1:31	0.254	2.3	23.3	19.1		100-1,200**	2.0	8.0
5,600	6,600	16	H 1:46	0.221	2.3	21	17		100-1,200**	2.0	8.0
6,800	8,800	12	H 1:46	0.166	2.2	15.7	12.8		100-1,200**	2.0	8.0
6,800	13,000	8	H 1:46	0.110	2.2	11	7		100-1,200**	2.0	6.5

48 V

Load max. (N)	Self-lock min. (N)	Pitch (mm/ spindle rev.)	Gear/ Ratio	Hall Resolution mm/count	End-play in mm	*Typical speed (mm/s)			Standard stroke lengths (mm) in steps of 50 mm	*Typical amp. (A)	
						No load	Full load	Reg- ula- ted		No load	Full Load
500***	1,000	20	E 1:7	0.721	4.5	160	141	120	100-1,200	1.5	7
1,700	2,200	20	F 1:18	0.721	3.5	71.2	59.0	55	100-1,200	1.5	7
2,000	2,600	16	F 1:18	0.577	2.9	57.2	35.0	44	100-1,200	1.5	7
2,600	3,400	12	F 1:18	0.433	2.6	42.9	35.0	33	100-1,200	1.5	7
3,400	4,400	16	G 1:31	0.339	2.4	32.5	27.0	25.8	100-1,200**	1.5	7
4,500	5,800	12	G 1:31	0.254	2.3	25.7	20.0	19.3	100-1,200**	1.5	7
5,600	6,600	16	H 1:46	0.221	2.3	21.9	18.0	16.8	100-1,200**	1.5	7
6,800	8,800	12	H 1:46	0.166	2.2	17.4	15.0	12.6	100-1,200**	1.5	7
6,800	13,000	8	H 1:46	0.110	2.2	11.5	9.0	4.8	100-1,200**	1.5	5.5

* The typical values can have a variation of $\pm 20\%$ on the current values and $\pm 10\%$ on the speed values. Measurements are made with an actuator in connection with a stable power supply and an ambient temperature at 20°C.

** There are limitations on the stroke length. If you need full load, please see: "Load v. Stroke Length".

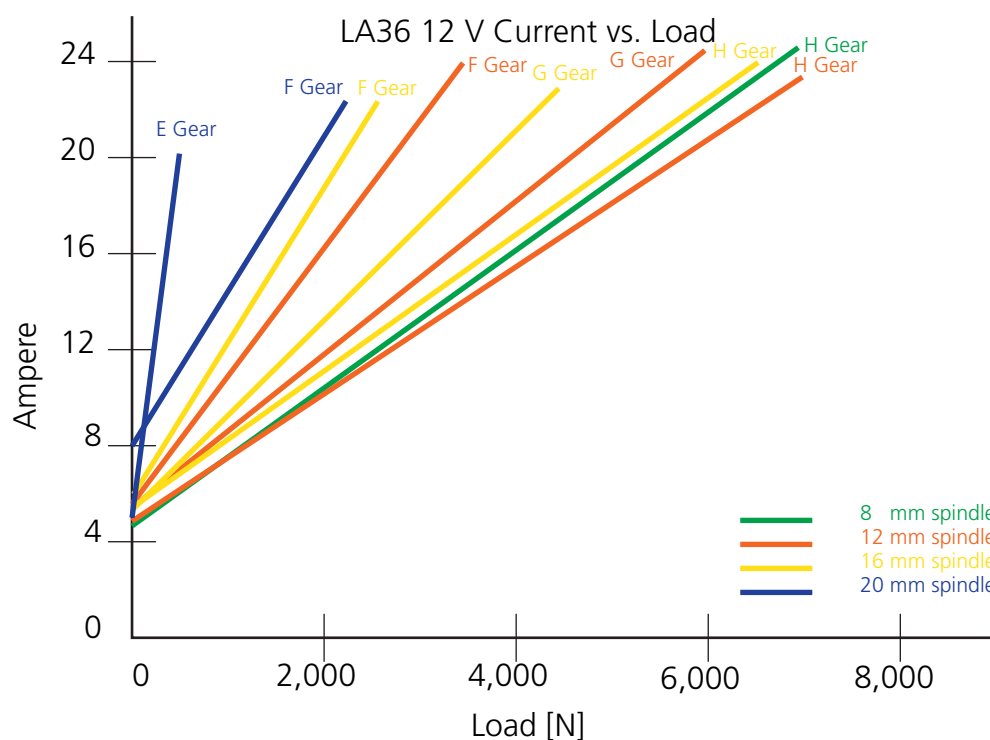
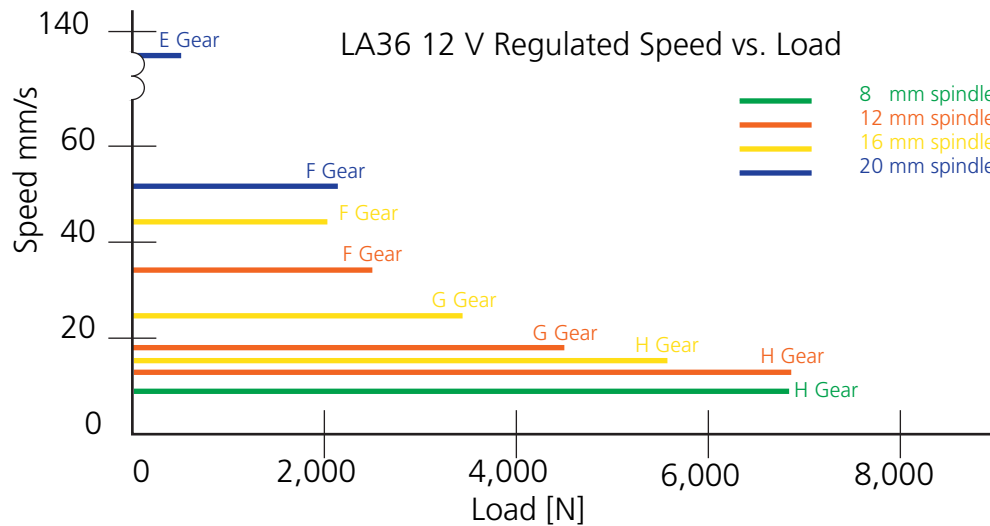
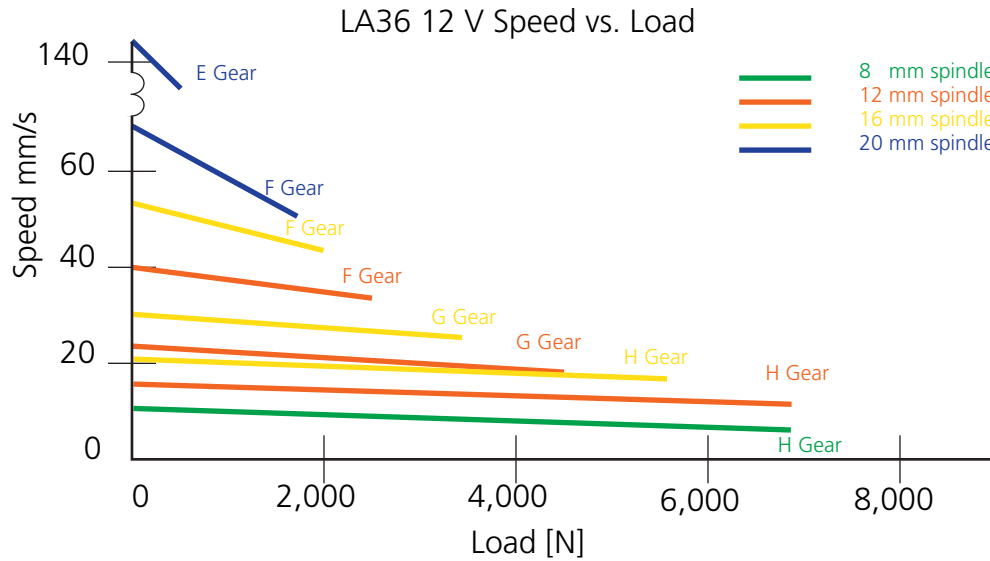
*** Note: Fully loaded actuators need a soft start in order to prevent the clutch from slipping when starting (see curves).



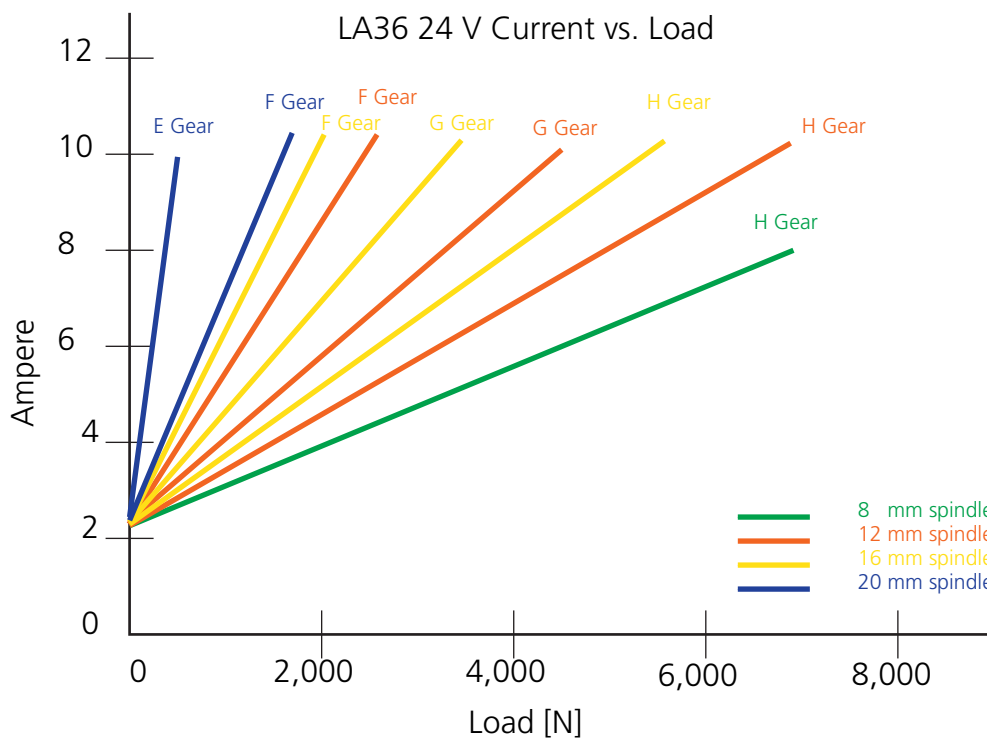
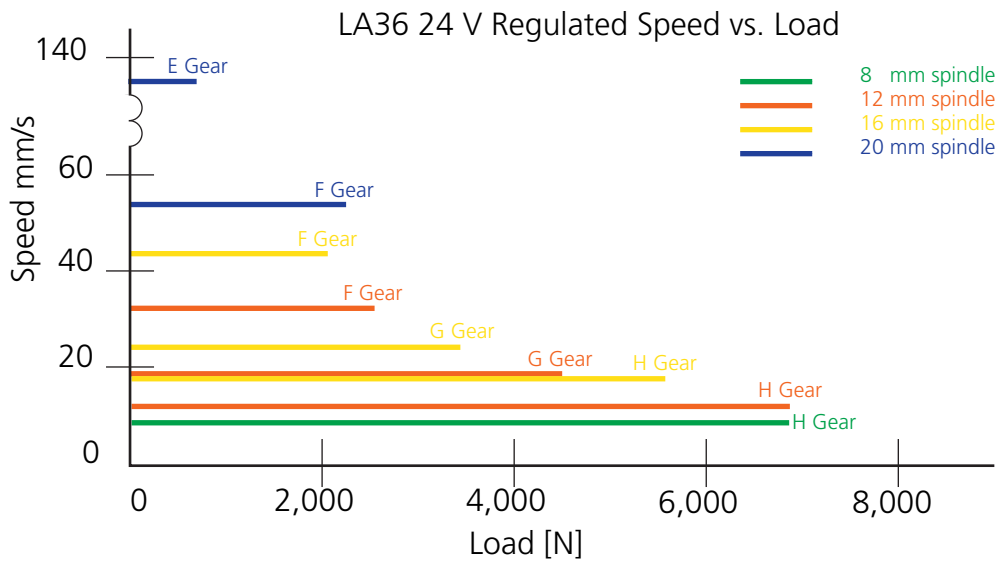
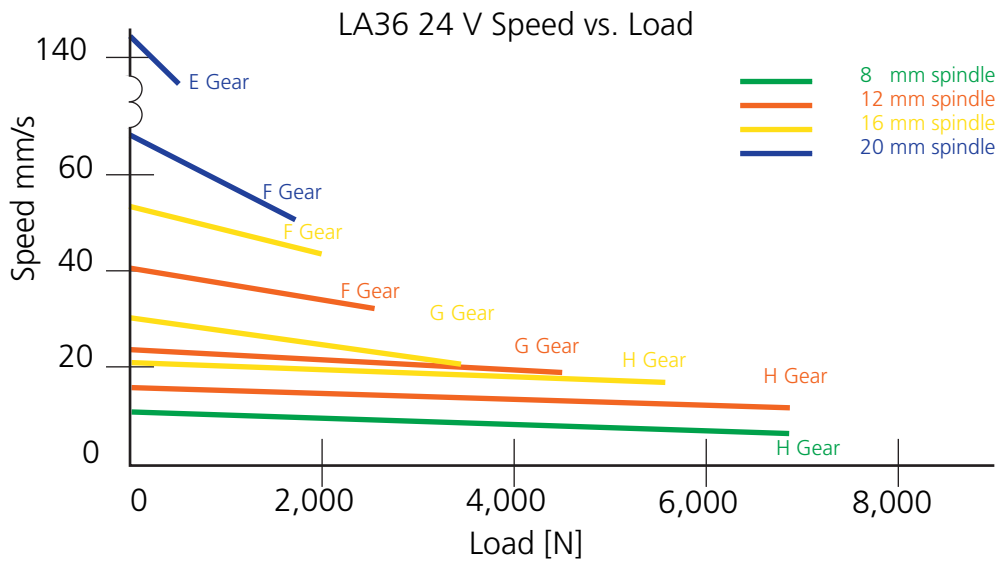
Please note that all actuators featuring "LIN bus", "CAN SAE J1939", "CANopen" and "Modbus" will run at a regulated speed, which is typically around 80% of the speed at the nominal load.

Speed and current curves:

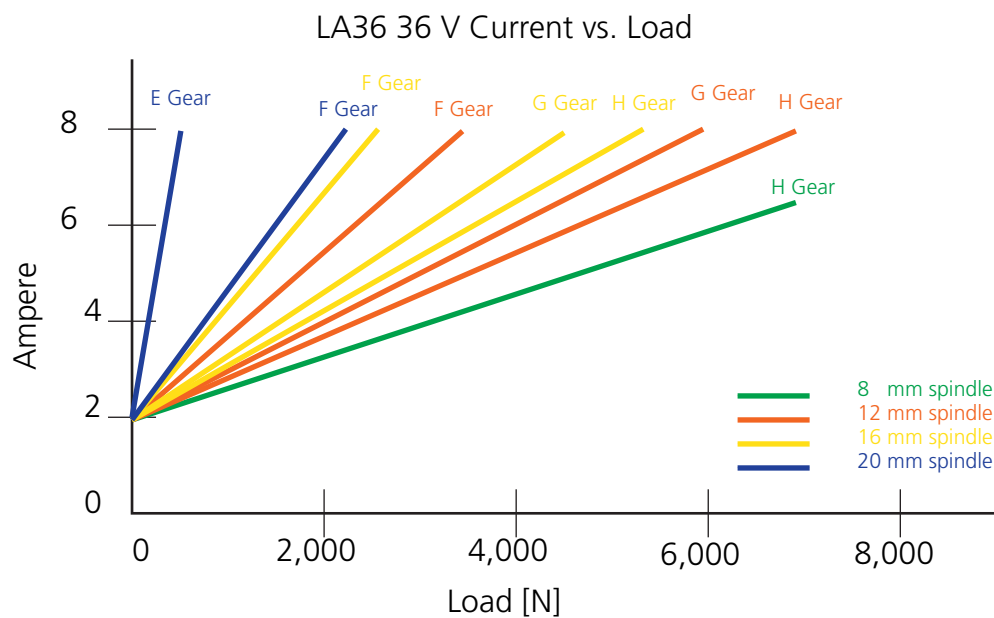
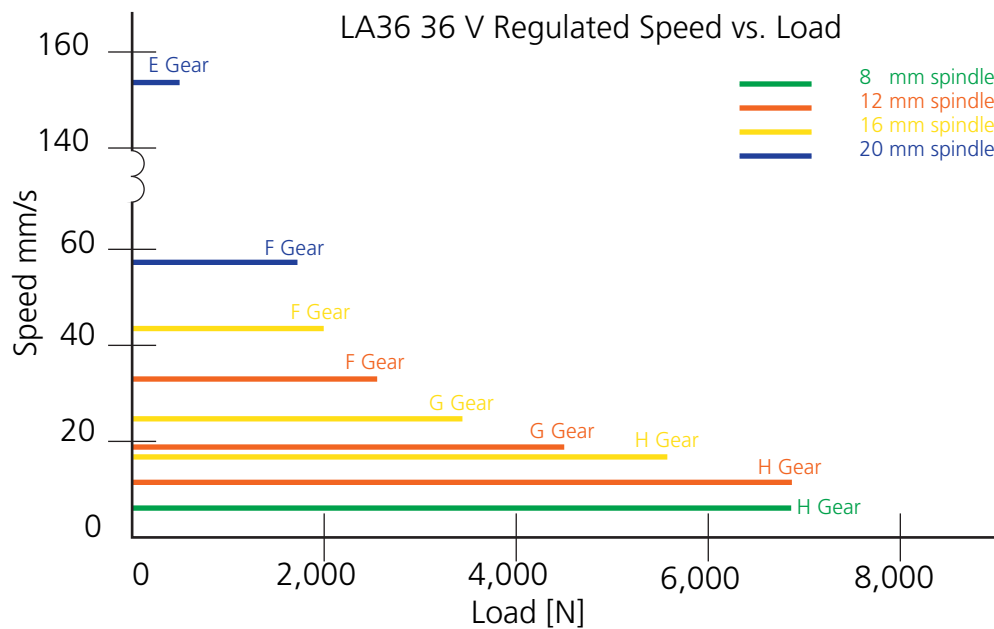
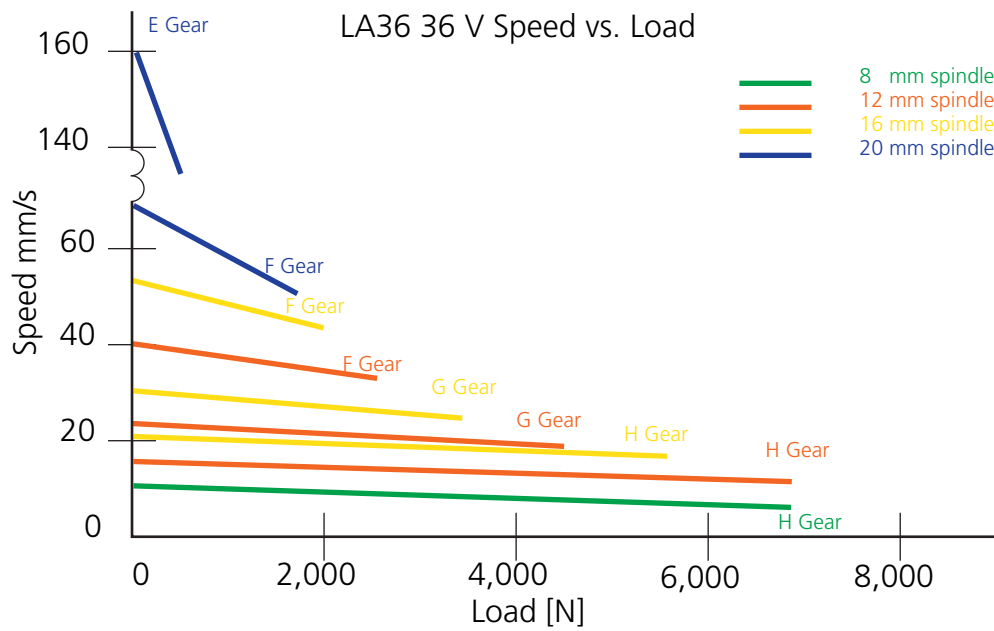
The typical values below are made with a stable power supply and an ambient temperature of 20°C.



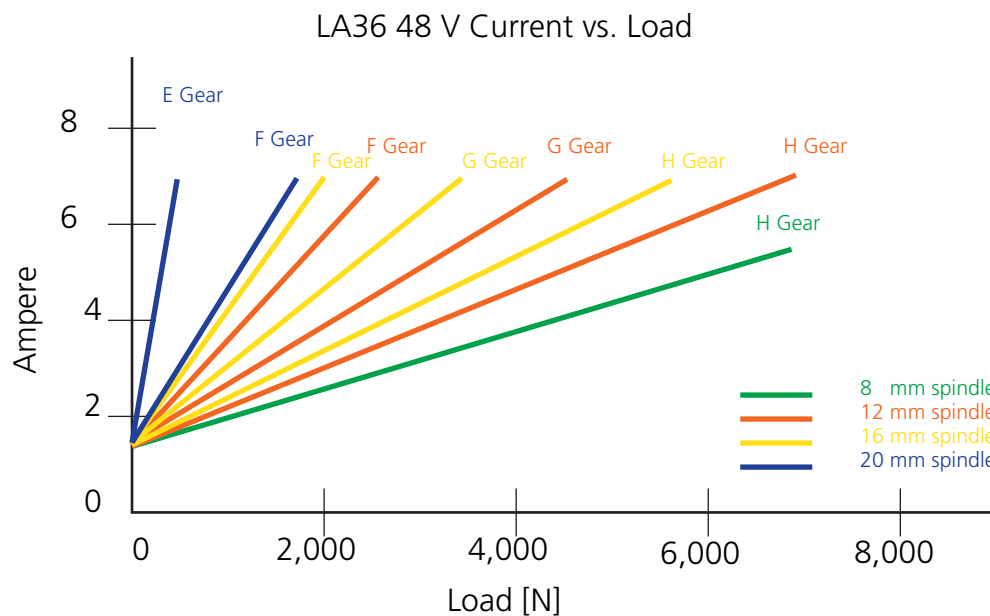
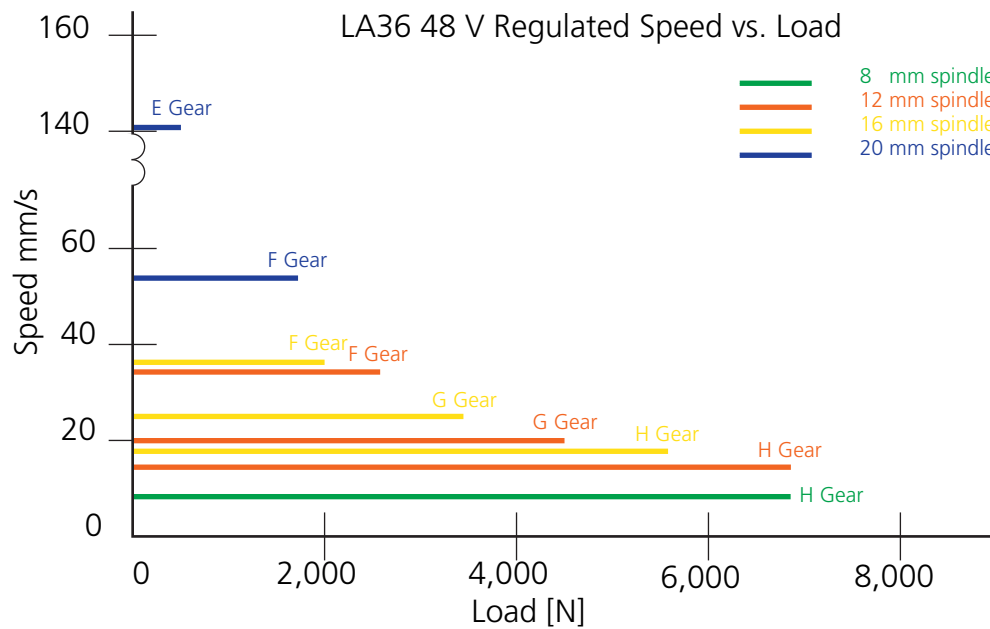
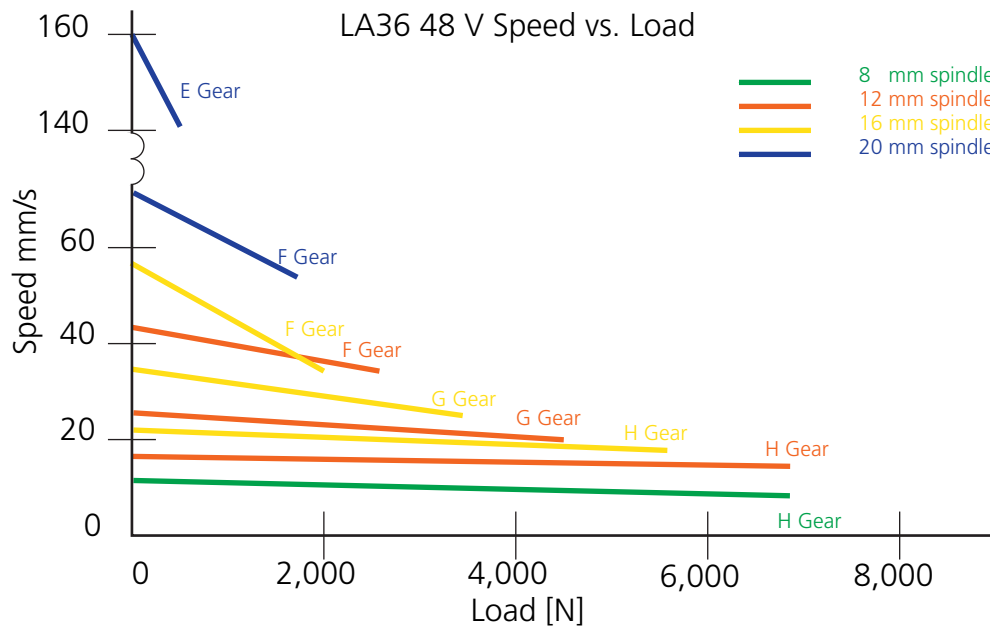
The typical values below are made with a stable power supply and an ambient temperature of 20°C.



The typical values below are made with a stable power supply and an ambient temperature of 20°C.



The typical values below are made with a stable power supply and an ambient temperature of 20°C.



Current limits

As described in the algorithm on previous page

Platform		12 V	24 V	48 V	Reference temperature: 0°C
B3 C3 F3	I/O Basic I/O Customised I/O Full	26 A	13 A	8 A	Above
		26 A	26 A	13 A	Below
0B	IO-Link	-	16 A	-	Above
		-	26 A	-	Below
14	Modbus RTU	-	16 A	8 A	Above
		-	26 A	15 A	Below
A7 A8	CANbus J1939 CANopen	-	13 A	8 A	Above
		-	26 A	13 A	Below
0E 2E 4E	Modbus TCP/IP Ethernet PROFINET	-	16 A	8 A	Above
		-	26 A	16 A	Below
C6 D6 E6	Off-highway: LIN bus CAN SAE J1939 CANopen	26 A	13 A	-	Above
		26 A	26 A	-	Below

Max. Current

The current in not limited by the actuator below is the anticipated consumption at max. load.
See: Recommended fuse for actuators without Integrated Controller

Platform		12 V	24 V	36 V	48 V	Reference temperature: 0°C
00 01	Standard Standard with power switch	26 A	13 A	10 A	8 A	Above
		26 A	13 A	10 A	8 A	Below

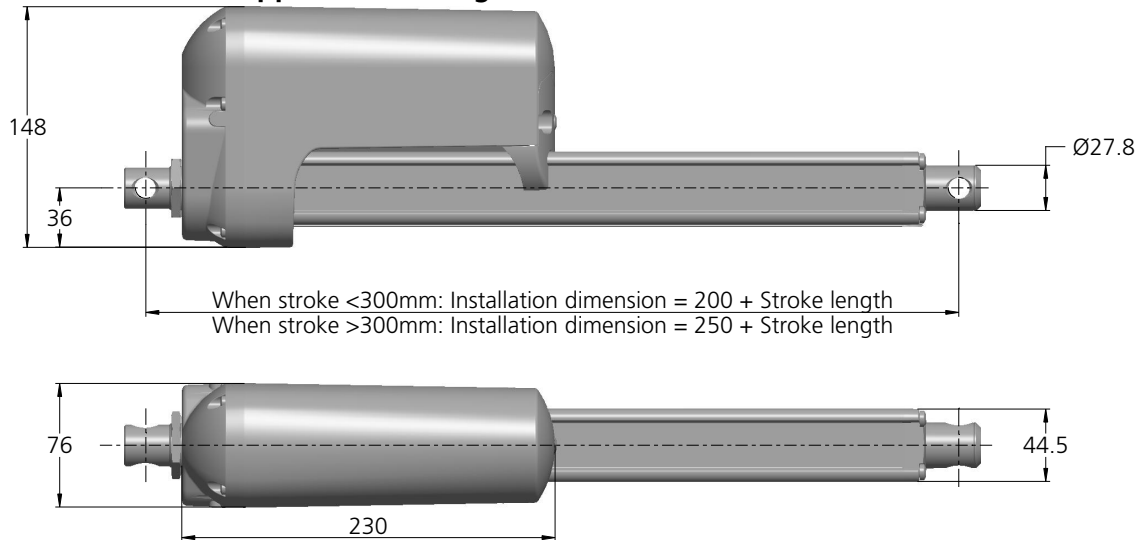
Current cut-offs

The principle behind the current cut-off measurement is an 'above limit' and a 'below limit' accumulating counter. When the time-out counter reaches a specific value the current cut-off goes into effect. The time-out value is pre-set at 200 ms.

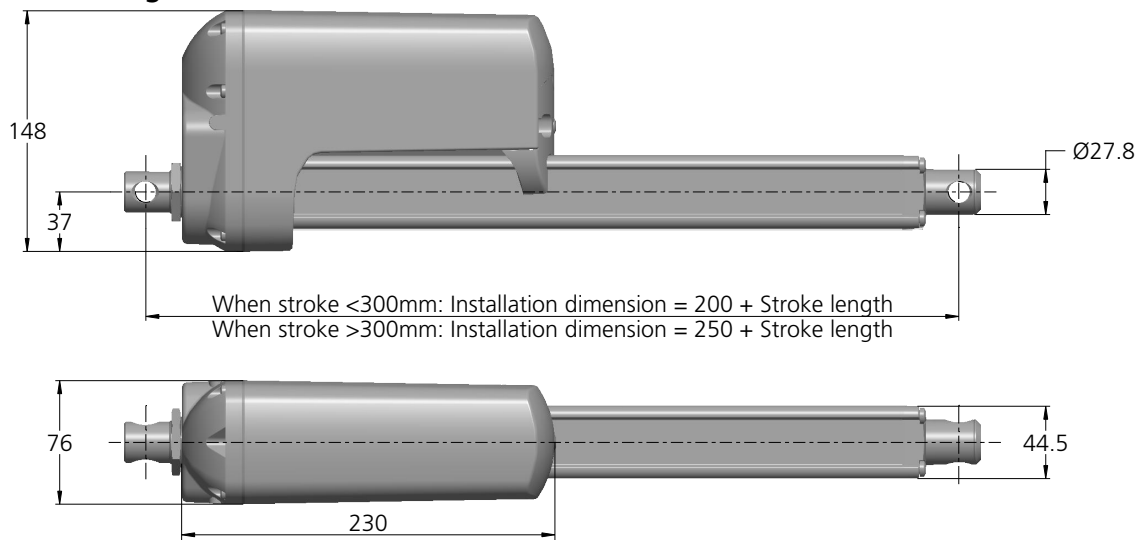
Platform		12 V	24 V	48 V	Reference temperature: 0°C
16	LIN bus	30 A	-	-	Above
		30 A	-	-	Below
07 08	CAN SAE J1939 CANopen	30 A	20 A	13 A	Above
		30 A	25 A	15 A	Below
17 18	CAN SAE J1939 CANopen	30 A	20 A	-	Above
		30 A	25 A	-	Below

Built-in dimensions

IP66, ATEX / IECEx / CCC approved housing

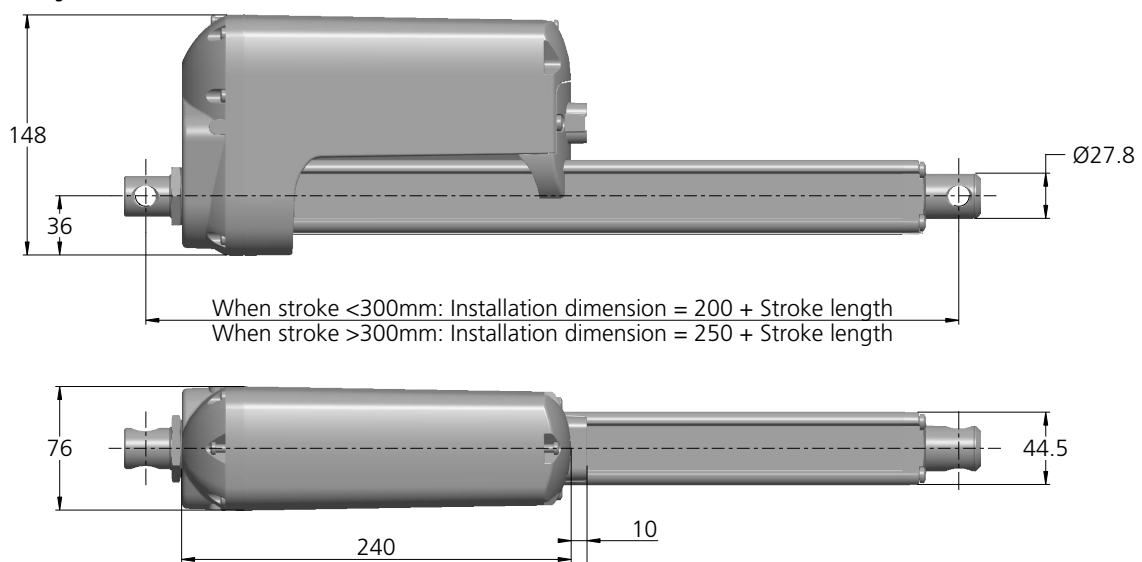


Reinforced housing



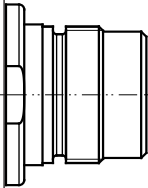
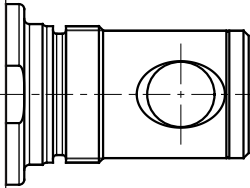
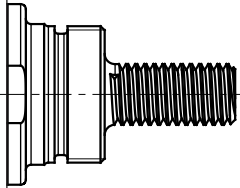
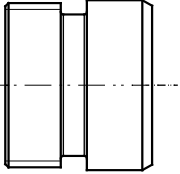
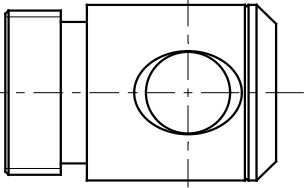
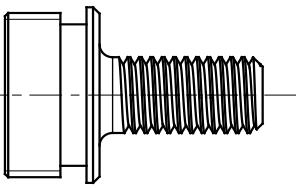
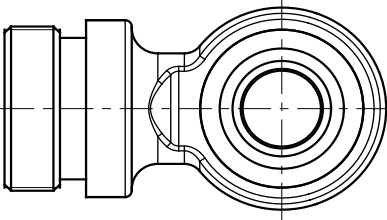
Minimum built-in dimension is 300 mm

Off-highway



Built-in dimensions with different PRE and BF

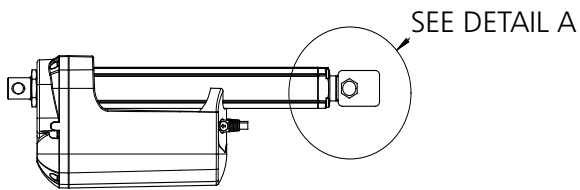
All dimensions are in mm

	Back fixture						
Length of stroke		≤300	>300	≤300	>300	≤300	>300
Piston rod eye		Inner thread - from the surface		Solid or slotted fixture - to center of the hole		Outer thread - from the surface	
	Inner thread - from the surface	189	239	195	245	180	230
	Solid or slotted fixture - to center of the hole	194	244	200	250	185	235
	Outer thread - from the surface	181	231	187	237	173	223
	Ball eye - to center of the hole	209	259	215	265	200	250

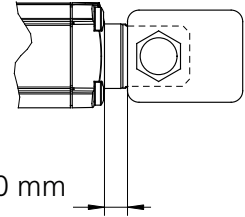
Keep a clearance when mounting a bracket



When mounting a custom bracket on the moving part of the actuator, please observe the minimum clearance between bracket and cylinder top when fully retracted. This will prevent jamming and destruction of the actuator drive train.



DETAIL A



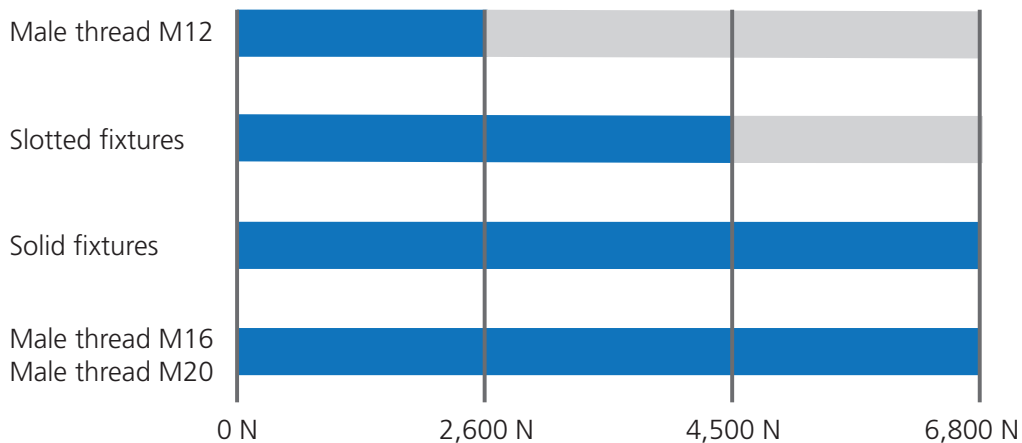
Min. clearance 2.0 mm

With Zero point the minimum stroke is 100 mm

The Zero point initialisation zone is located between 35-75 mm going from the most inward position.

The movement passing the zone has to be stable for the initialisation to succeed - also no virtual limits can be set in the initialisation zone.

Durability for piston rod eyes and back fixtures



Blue = Full Lifetime

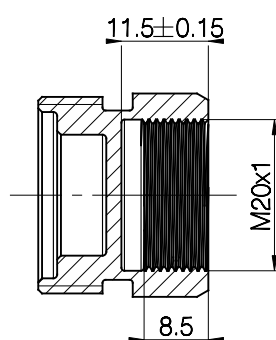
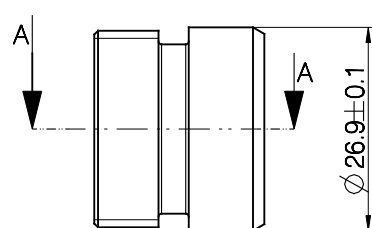
Grey = Reduced lifetime

(If e.g. a Male thread M12 is used with an actuator with a larger load than 2,600 N and a Slotted fixture is used with an actuator with a larger load than 4,500 N their lifetime will be shorter than if the other fixtures are chosen).

Piston rod eyes

LINAK P/N: 0361016

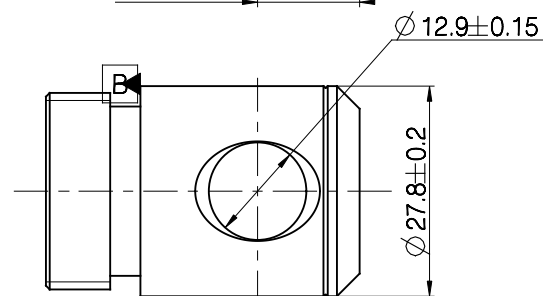
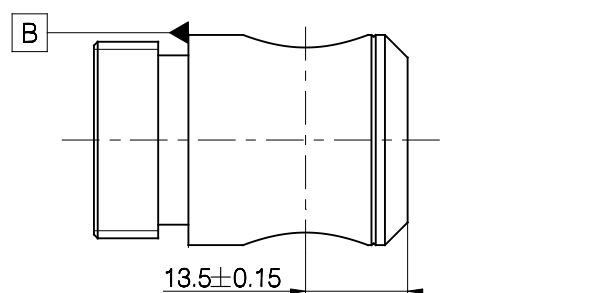
AISI 303



SECTION A-A

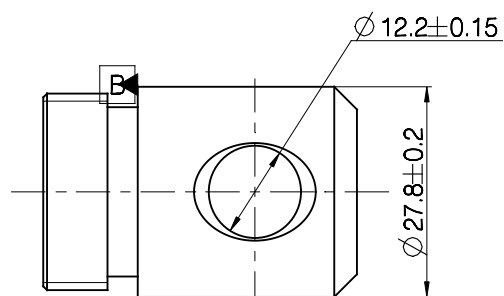
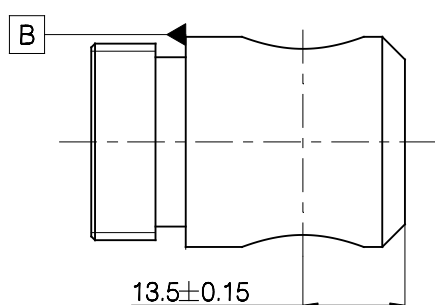
LINAK P/N: 0361018

Free-cutting steel with galvanised surface



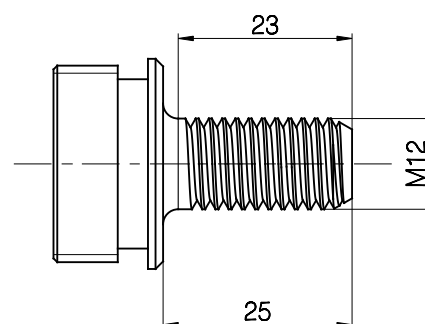
LINAK P/N: 0361109

Free-cutting steel with galvanised surface



LINAK P/N: 0361224

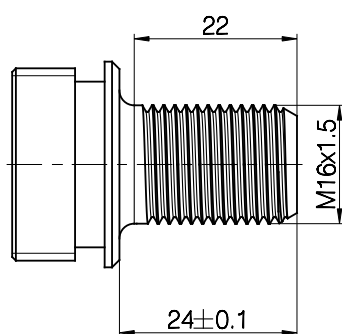
AISI 303



Piston rod eyes:

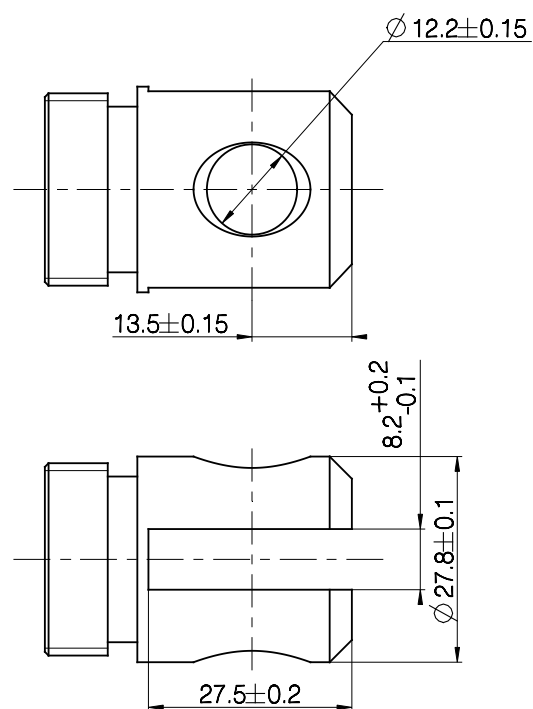
LINAK P/N: 0361135

AISI 303



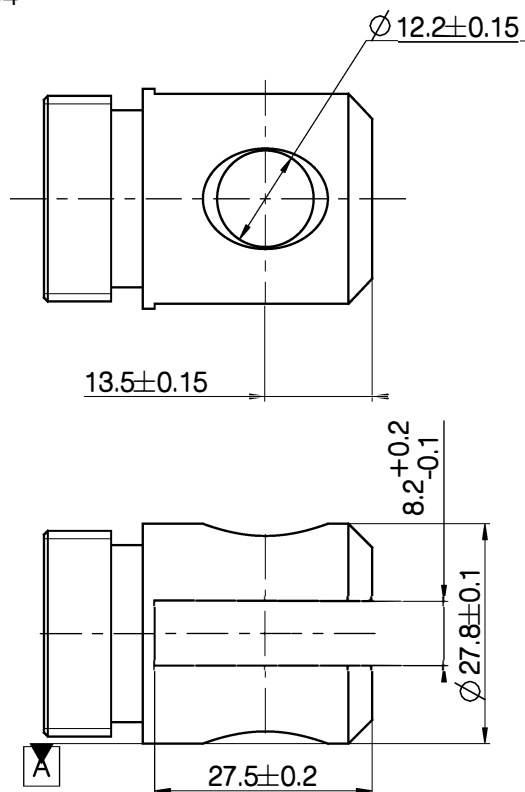
LINAK P/N: 0361138

Free cutting steel with galvanised surface.



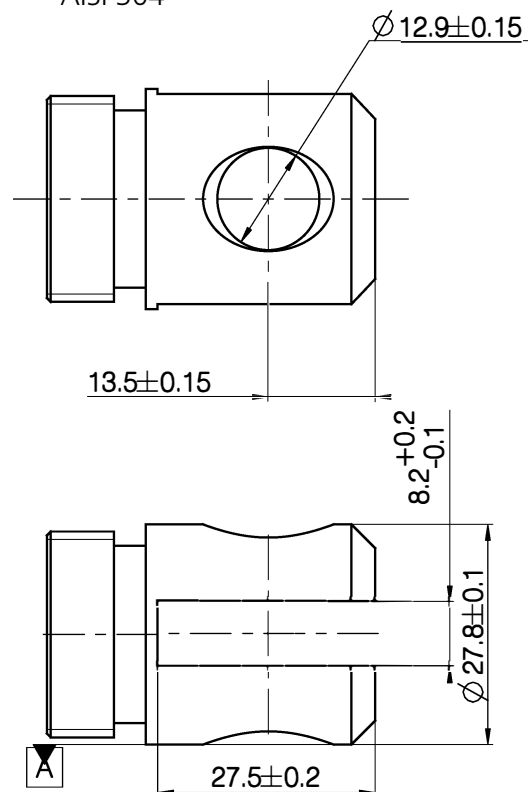
LINAK P/N: 0361260

AISI 304



LINAK P/N: 0361275

AISI 304

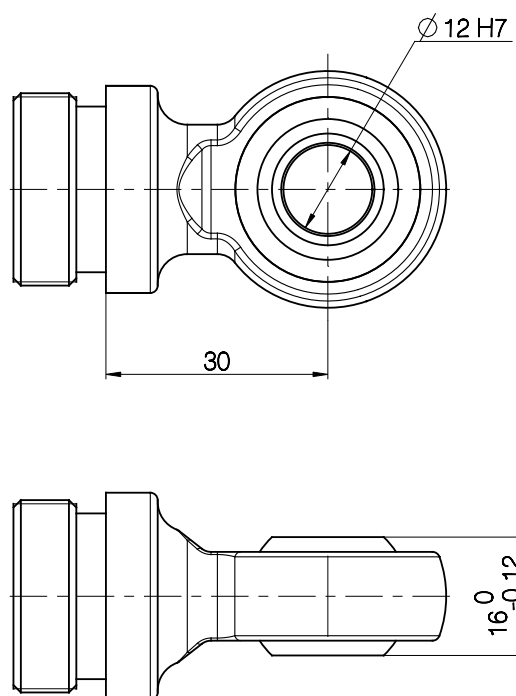


Piston rod eyes:

LINAK P/N: 0361350

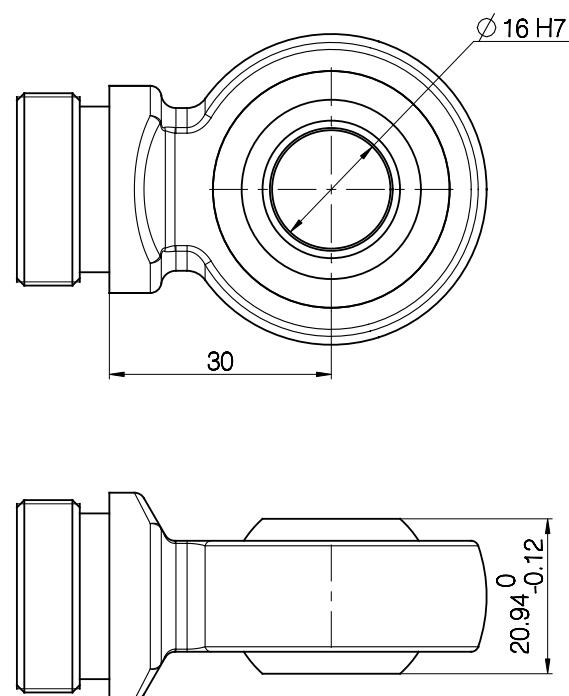
6.8 KN = Max. load 6.800 N in pull

AISI 304



LINAK P/N: 0361351

AISI 304

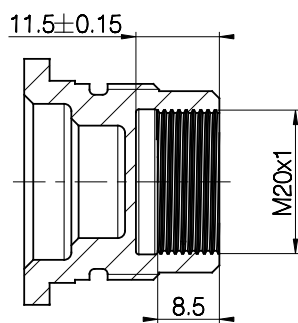
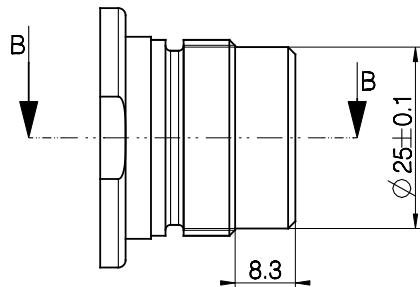


The piston rod eye is only allowed to turn 0 - 180 degrees.

Back fixtures:

LINAK P/N: 0361761

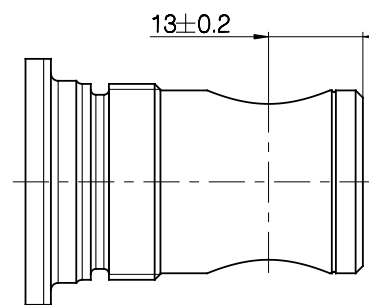
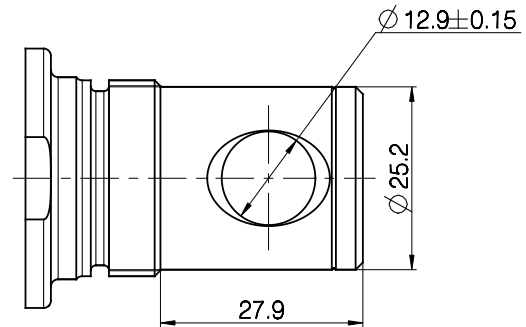
AISI 303



SECTION B-B

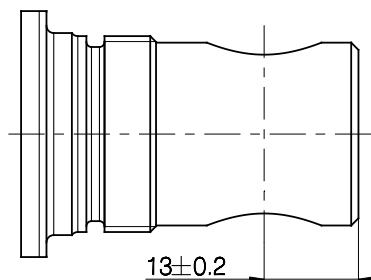
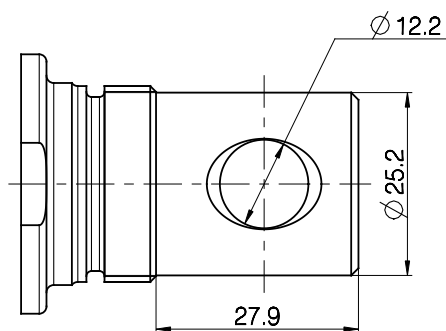
LINAK P/N: 0361715

Free cutting steel with galvanised surface.



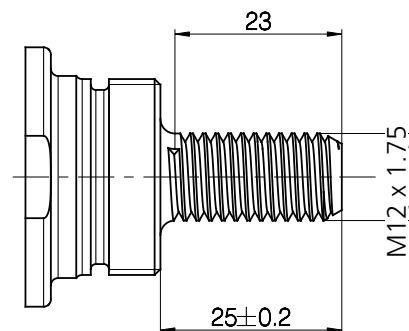
LINAK P/N: 0361714

Free cutting steel with galvanised surface.



LINAK P/N: 0361753

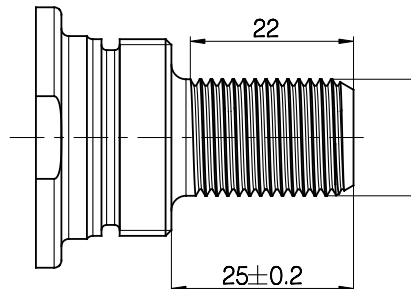
AISI 303



Back fixtures:

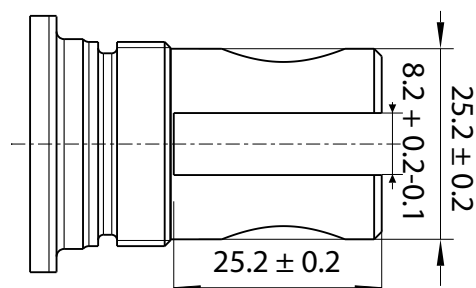
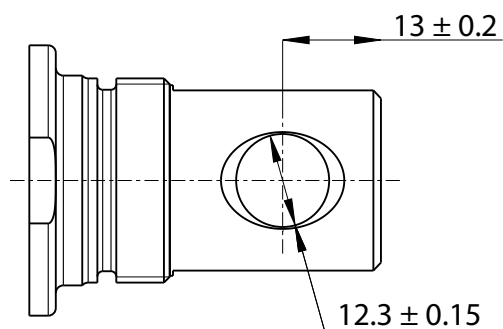
LINAK P/N: 0361754

AISI 303



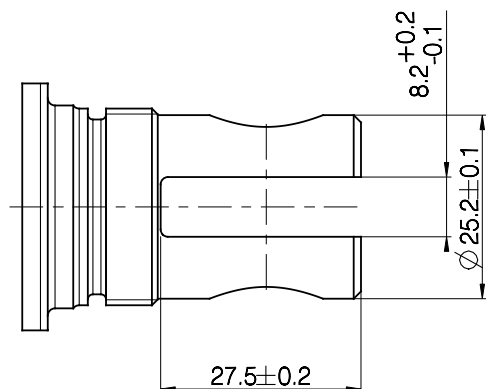
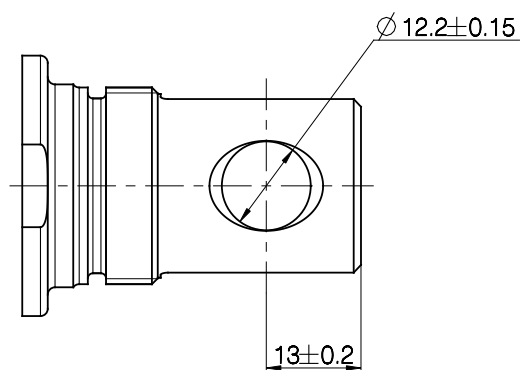
LINAK P/N: 0361713

Free cutting steel with galvanised surface.



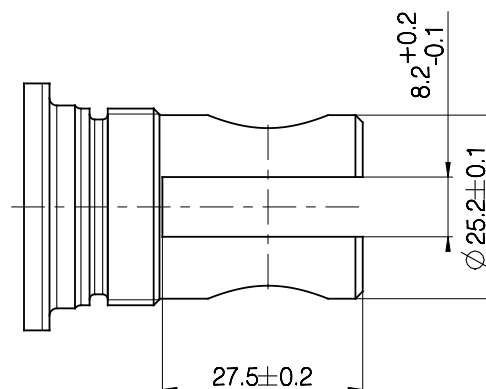
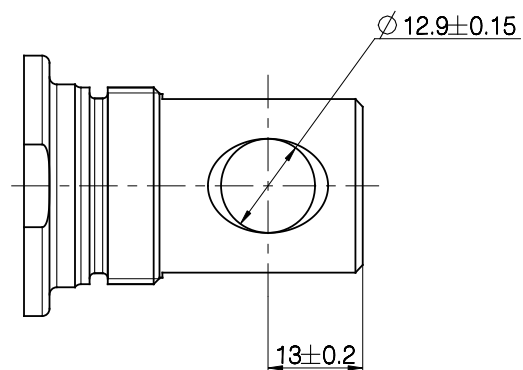
LINAK P/N: 0361742

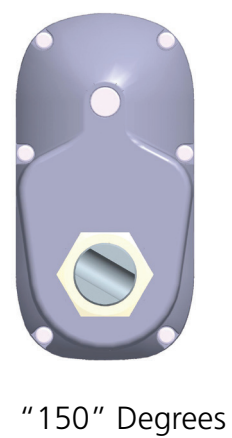
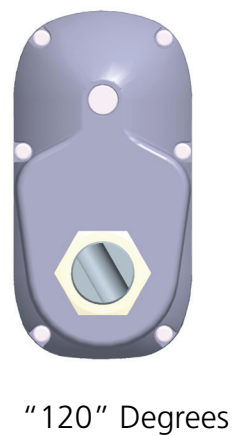
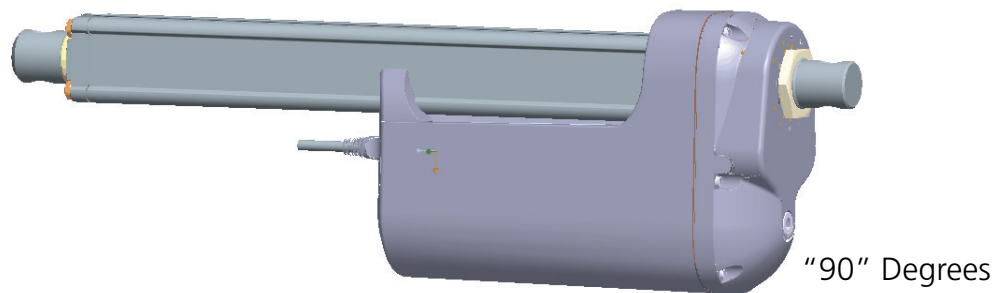
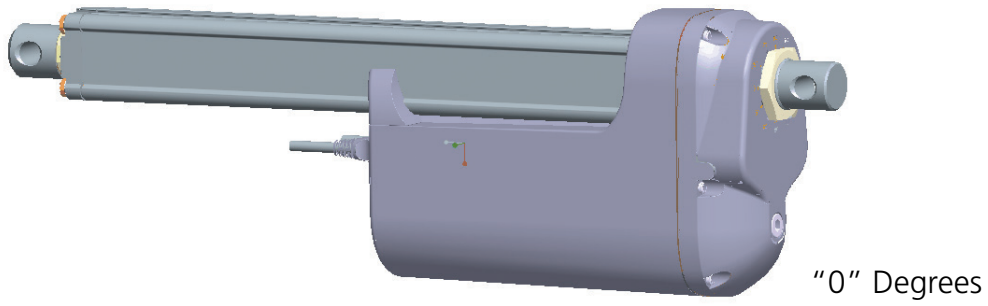
AISI 304



LINAK P/N: 0361743

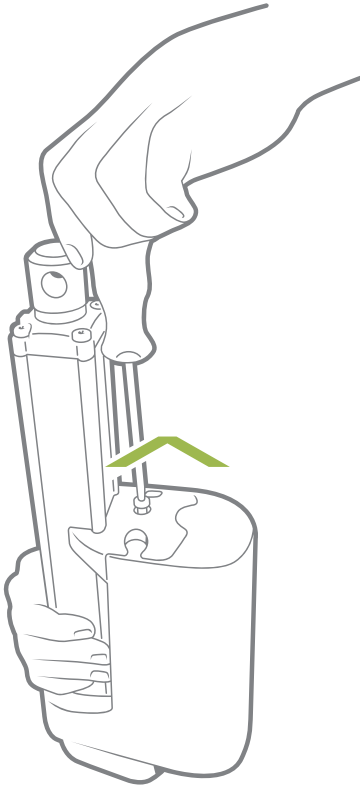
AISI 304



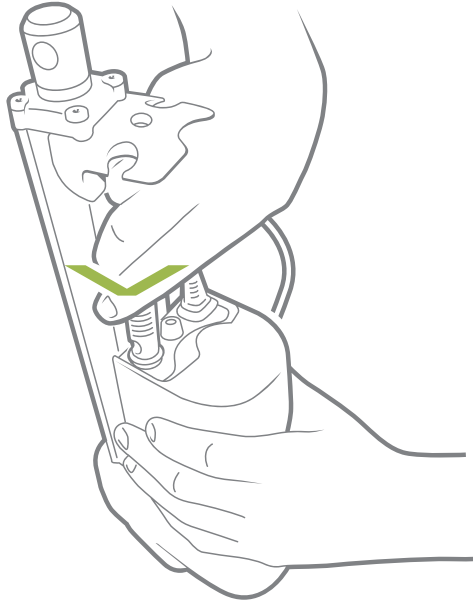
Back fixture orientation:

NB. All with tolerance of $\pm 4^\circ$

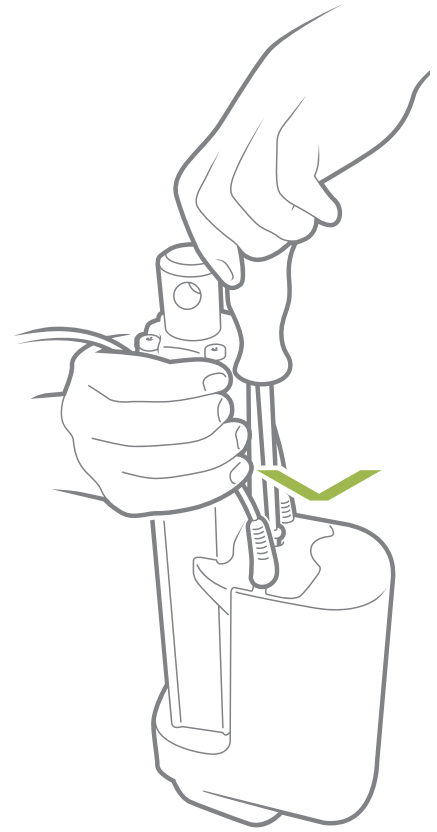
Mounting of cables



1. Unscrew the screw and separate the cover from the housing. Remove the blind plug(s).



2. Plug in the power cable and/or the signal cable.



3. Slide the cover onto the actuator.
The torque of the cover screw is approx. 3.5 ± 0.3 Nm
TORX 25IP



When changing the cables on a LINAK® actuator, it is important that this is done carefully, in order to protect the plugs and pins. Before the new cable is mounted, we recommend that the socket is greased with vaseline, to keep the high IP protection and ensure an easy mounting. Please be sure that the plug is in the right location and fully pressed in before the cable lid is mounted.

Please note that if the cables are mounted and dismantled more than 3 times the plugs can be damaged. Therefore, we recommend that such cables are discarded and replaced.

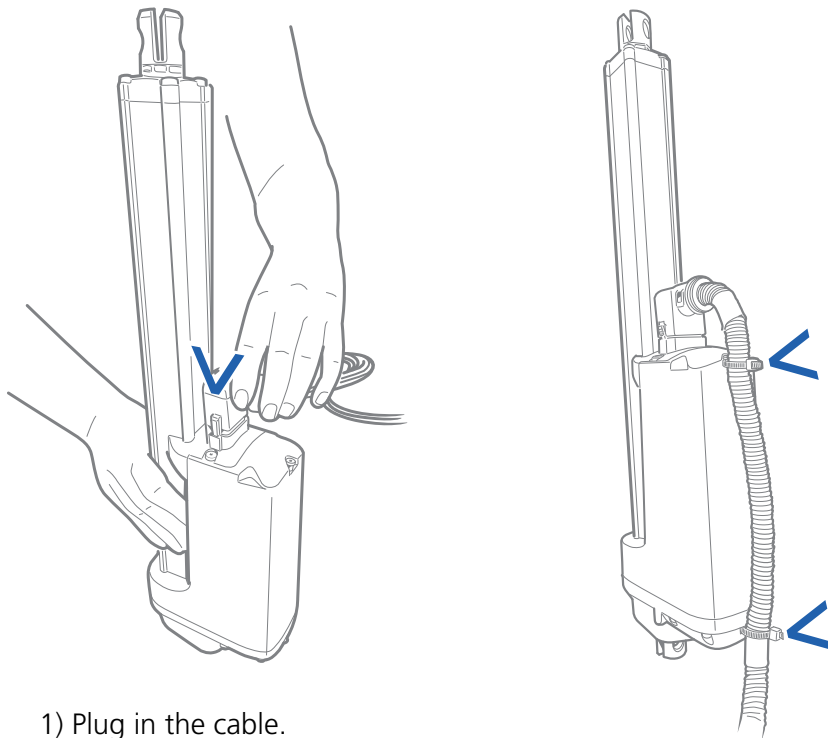
Also note that the cables should not be used for carrying the actuator.

We recommend to take some precaution and design the wire connection in a way, where the cable end is kept inside a closed, protected area to guarantee the high IP protection.



NOT valid for ATEX cables, please refer to the ATEX section for correct cable mounting on ATEX actuators.

Mounting of cable with Off-highway



1) Plug in the cable.

An audible "Click" confirms a correct mounting

2) Secure the cable with cable-ties to the two anchors



We recommend to take some precaution and design the wire connection in a way, where the cable end is kept inside a closed, protected area to guarantee the high IP protection.

Off-highway connection to Actuator Connect™

When connecting the actuator to Actuator Connect™ it is imperative to follow these instructions.

Power supply connection:

The actuator has to be powered with 12 Volt DC

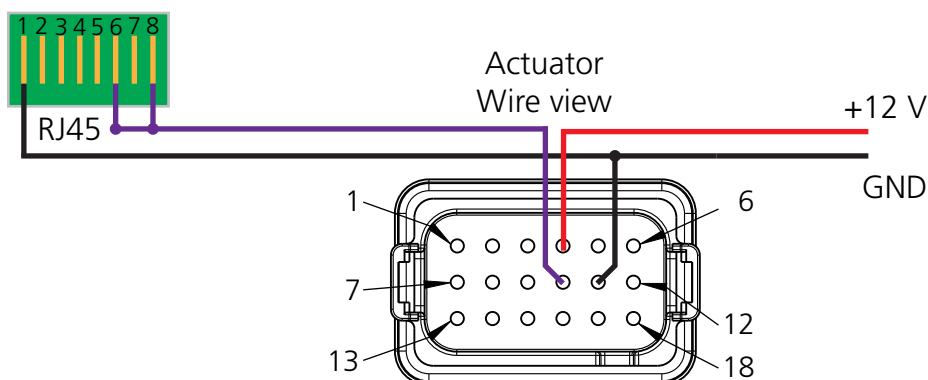
Positive is connected to pin 4 at the actuator

GND is connected to pin 11 at the actuator

RJ45:

Pin 1 at the RJ45 plug is Ground and has to be interconnected to GND on pin 11 at the actuator

Pin 6 and 8 at the RJ45 plug is communication and both has to be connected to pin 10 at the actuator:

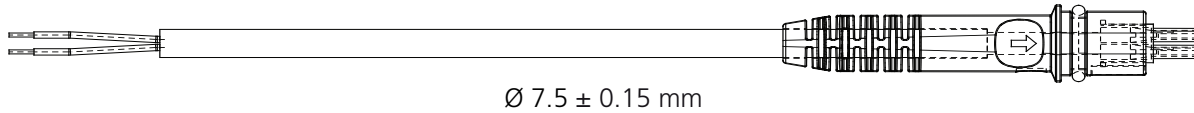


Cables

Power cable dimensions

LINAK® P/N 0367046

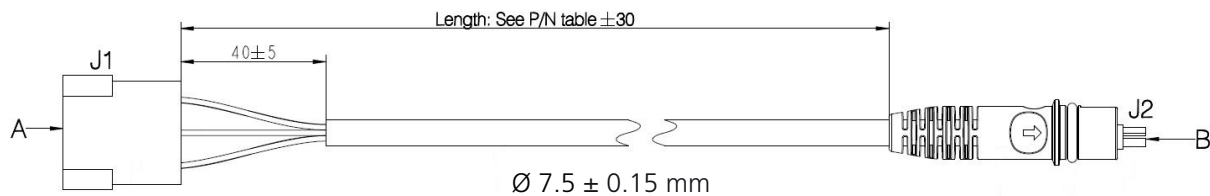
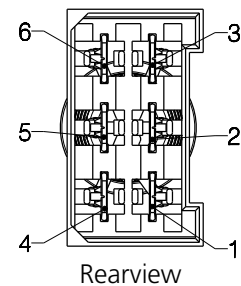
Colour	Outer dimensions	Core mm ²	AWG*
Brown	Ø2.8 mm	2.0	14
Blue	Ø2.8 mm	2.0	14



6-pin Signal cable dimensions

LINAK P/N 0367049

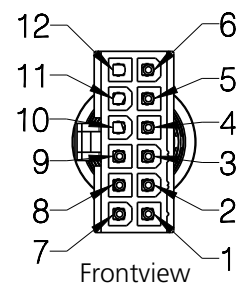
Colour	Outer dimensions	Core mm ²	AWG*
Violet	Ø1.5 mm	0.5	20
Black	Ø1.5 mm	0.5	20
Red	Ø1.5 mm	0.5	20
Yellow	Ø1.5 mm	0.5	20
Green	Ø1.5 mm	0.5	20
White	Ø1.5 mm	0.5	20



9-pin Signal cable dimensions

LINAK P/N 0368543

Colour	Outer dimensions	Core mm ²	AWG*	Pin
Orange	Ø1.5 mm	0.5	20	5
Black	Ø1.5 mm	0.5	20	1
Red	Ø1.5 mm	0.5	20	2
Light Blue	Ø1.5 mm	0.5	20	6
Yellow	Ø1.5 mm	0.5	20	3
Green	Ø1.5 mm	0.5	20	4
Grey	Ø1.5 mm	0.5	20	0
Violet	Ø1.5 mm	0.5	20	7
White	Ø1.5 mm	0.5	20	8



*AWG: American Wire Gauge

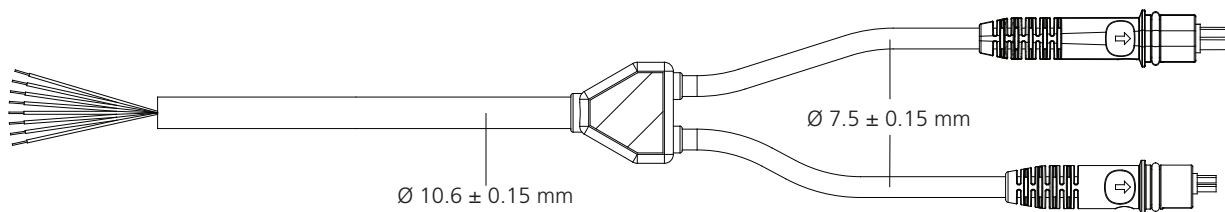
Cables

Y-cable dimensions

LINAK® P/N 0367020

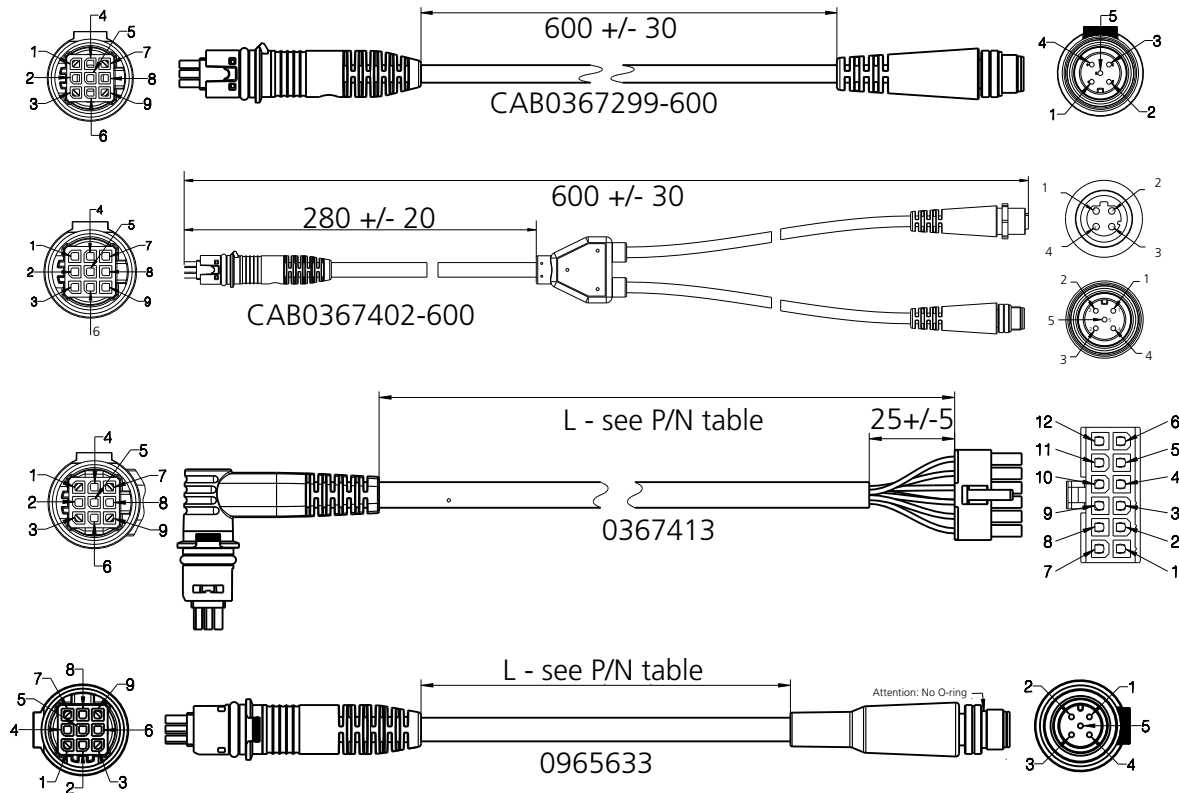
Colour	Outer dimensions	Core mm ²	AWG*	Pin**
Brown	Ø2.8 mm	2.0	14	2
Blue	Ø2.8 mm	2.0	14	1
Red	Ø1.5 mm	0.5	20	4
Black	Ø1.5 mm	0.5	20	3
Yellow	Ø1.5 mm	0.5	20	7
Green	Ø1.5 mm	0.5	20	8
White	Ø1.5 mm	0.5	20	5
Violet	Ø1.5 mm	0.5	20	6

** Pin connections are the same for both AMP and Deutsch connectors



Cable P/N Table					
LINAK P/N	Cable type	# Wires	mm ²	AWG*	Length in mm
0367006	Power cable with AMP	2	2.0	14	200
CAB0367046-0400	Power cable	2	2.0	14	400
CAB0367046-0600	Power cable	2	2.0	14	600
CAB0367046-1500	Power cable	2	2.0	14	1,500
CAB0367046-5000	Power cable	2	2.0	14	5,000
CAB0367049-0600	Signal cable	6	0.5	20	600
CAB0367049-1500	Signal cable	6	0.5	20	1,500
CAB0367049-2000	Signal cable	6	0.5	20	2,000
CAB0367049-3000	Signal cable	6	0.5	20	3,000
CAB0367049-5000	Signal cable	6	0.5	20	5,000
CAB0368543-1500	Signal cable	9	0.5	20	1,500
CAB0368543-5000	Signal cable	9	0.5	20	5,000
CAB0367299-600	Signal cable for IO-Link	9	0.5	20	600
CAB0367402-600	Signal Y-cable for Ethernet	9	0.5	20	600
0367413-XXXX	Signal cable w. 90° plug	9	0.5	20	600 1,500 1,000
0965633-1000	Signal cable for Modbus RTU	9	0.5	20	1000
CAB0367020-XXXX	Y-Cable Signal and Power	6	0.5	20	1,500 5,000
		2	2.0	14	
CAB0367430-XXXX	Y-Cable Signal and Power	9	0.5	20	1,500 5,000 10,000
		2	2.0	14	
0367440-XXXX	Y-Cable w. 90° plug Signal and Power	9	0.5	20	1,500 5,000 10,000
		2	2.0	14	

*AWG: American Wire Gauge



Cable kit article numbers

BusLink cable kits					
Platform		Article no.	Connection	Includes	Colour
04	Modbus (IC) - platform no longer available	0367998	RJ45	1. Adapter 2. USB2LIN cable	Yellow
07	CAN SAE J1939	0367997	RJ45	(Adapter + USB2Lin)	Green
08	CANopen				
16*	LIN bus				
17*	CAN SAE J1939				
18*	CANopen				

Actuator Connect™ cable kits					
Platform		Article no.	Pins	Includes	Colour
B3	I/O Basic	0367996	Signal-power + RJ45	(Adapter + USB2Lin)	Grey
C3	I/O Customised				
F3	I/O Full				
B7	CAN SAE J1939				
B8	CANopen				
0B	IO-Link				
A7	CAN SAE J1939				
A8	CANopen				
2E	EtherNet/IP				
0E	Modbus TCP/IP				
14	Modbus RTU				
4E	PROFINET				



Latest versions of both BusLink® and Actuator Connect® can be downloaded at the [LINAK/TECHLINE](#) page.

* Utilises Buslink cable -but are configured and monitored with Actuator Connect®.

Electrical installation



- To ensure maximum self-locking ability, please be sure that the motor is shorted when stopped. Actuators with integrated controller provide this feature, as long as the actuator is powered.
- When using soft stop on a DC-motor, a short peak of higher voltage will be sent back towards the power supply. When selecting the power supply, it is important to make sure that it does not turn off the output when this backwards load dump occurs.



The power supply for actuators without integrated controller must be monitored externally and cut off in case of current overload. IC actuators have an integrated overcurrent protection.

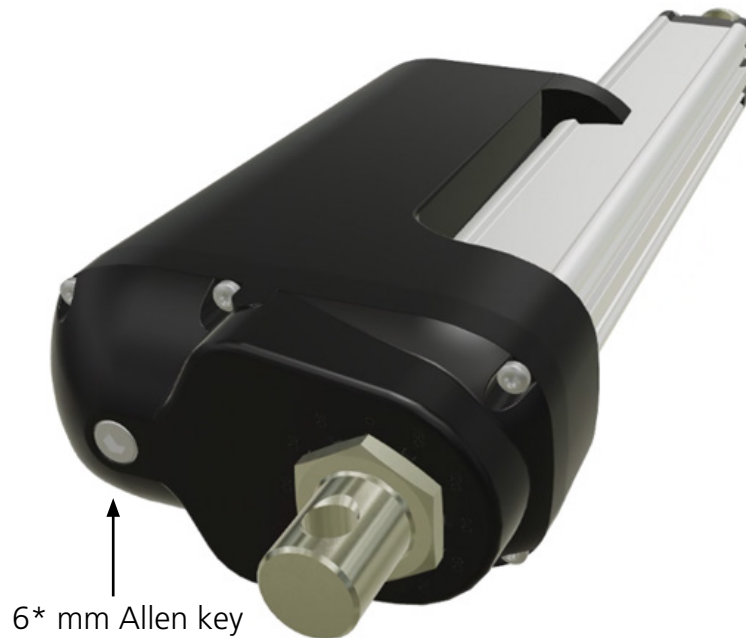
Manual hand crank:

The manual Hand Crank can be used in the case of a power failure *-it is only intended for emergency use.*



The manual hand crank must **NEVER** be used in an ATEX/IECEX/CCC environment, as it is not possible to verify that the actuator is properly sealed after the hand crank has been used.

The actuator must **NOT** be used in an ATEX/IECEX/CCC environment, if the warning sticker is missing or shows sign of having been removed from the bottom of the actuator.



The cover over the allen key socket must be unscrewed before the allen key can be inserted and the hand crank operated.

Hand crank torque: 6-8 Nm

Hand crank rpm: Max. 65

Piston rod movement per hand crank turn:				
Spindle type:	8 mm	12 mm	16 mm	20 mm
Gear F:	-	11 mm	14 mm	18 mm
Gear G:	-	6 mm	8 mm	10 mm
Gear H:	3 mm	4 mm	5 mm	7 mm
Gear E:	-	-	-	27 mm

* 5 mm Allen key -if stainless steel piston rod eye and back fixture are chosen



- The power supply has to be disconnected during manual operation
- If the actuator is operated as a hand crank, it must only be operated by hand, otherwise there is a potential risk of exceeding the 6-8 Nm, resulting in overloading and thereby damaging the actuator - do **NOT** use power tools for the hand crank!
- After using the hand crank, the ingress protection will be less than IP66 once the plug is removed
- After using the hand crank, always return the actuator to the most inward position. Failing to do so can damage the actuator and/or the application it is used for

Environmental tests – Climatic

Test	Specification	Comment
Cold Test	EN 60068-2-1 (Ab)	Storage at low temperature: Temperature: -40 °C Duration: 72 h Not connected Tested at room temperature.
	EN 60068-2-1 (Ad)	Storage at low temperature: Temperature: -30 °C Duration: 2 h Actuator is not activated/connected. Tested at low temperature.
Dry Heat	EN 60068-2-2 (Bb)	Storage at high temperature: Temperature: +90 °C Duration: 72 h Actuator is not activated/connected. Tested at room temperature
	EN 60068-2-2 (Bd)	Storage at high temperature: Temperature: +70 °C Duration: 1,000 h Actuator is not activated/connected Tested at high temperature. Operating at high temperature: Temperature: +60 °C Int. max. 17 % Duration: 700 h Actuator is activated Tested at high temperature.
Change of Temperature	EN 60068-2-14 (Na)	Rapid change of temperature: High temperature: +100 °C in 60 minutes. Low temperature: -30 °C in 60 minutes. Transition time: <10 seconds Duration: 100 cycles Actuator is not activated/connected. Tested at room temperature.
	EN 60068-2-14 (Nb)	Controlled change of temperature: Temperature change 5 °C pr. minute High temperature: +70 °C in 60 minutes. Low temperature: -30 °C in 30 minutes. 130 minutes pr. Cycle. Duration: 1,000 cycles (90 days) Actuator is not activated/connected. Tested at 250, 500 and 1,000 cycles at low and high temperatures.

Test	Specification	Comment
Damp Heat	EN 60068-2-30 (Db)	Damp heat, Cyclic: Relative humidity: 93-98 % High temperature: +55 °C in 12 hours Low temperature: +25 °C in 12 hours Duration: 21 cycles * 24 hours Actuator is not activated/connected. Tested within 1 hour after condensation, after upper temperature has been reached.
	EN 60068-2-3 (Ca)	Damp heat, Steady state: Relative humidity: 93-95 % Temperature: +40 ± 2 °C Duration: 56 days Actuator is not activated/connected. Tested within one hour after exposure.
Salt Mist	EN 60068-2-52 (Kb)	Salt spray test: Salt solution: 5 % sodium chloride (NaCl) 4 spraying periods, each of 2 hours. Humidity storage 7 days after each. Actuator not activated/connected. Exposure time: 500 hours
Degrees of Protection	EN 60529 – IP66	IP6X - Dust: Dust-tight, No ingress of dust. Actuator is not activated. IPX6 – Water: Ingress of water in quantities causing harmful effects is not allowed. Duration: 100 litres pr. minute in 3 minutes Actuator is not activated. IPX6 –Connected actuator: Actuator is driving out and in for 3 min. 100 (l/min) jet of water is placed at the wiper ring for 3 (min). IPX6 –Connected actuator and push 6,800 (N) Actuator is driving out and in for 3 min. and Push 6,800 (N) at the end-pos. 100 (l/min.) jet of water is placed at the wiper ring for 3 min.
	DIN40050 – IP69K	High pressure cleaner: Water temperature: +80 °C Water pressure: 80 bar Spray angle: 45 ° Spray distance: 100 mm Duration: From any direction 10 seconds of spraying followed by 10 seconds rest. Actuator is not activated. Ingress of water in quantities causing harmful effects is not allowed.
	DUNK test	The actuator has been warmed up to 115 °C for 20 hours. After this it is cooled down in 20 °C salt water. Cooling time: 5 minutes Opened for checking salt deposit and water.
Chemicals	BS7691 / 96 hours	Diesel 100 % Hydraulic oil 100 % Ethylene Glucol 50 % Urea Nitrogen saturated solution Liquid lime 10 % (Super- Cal) NPK Fertilizer (NPK 16-4-12) saturated Tested for corrosion.

Environmental tests - Mechanical

Test	Specification	Comment
Free Fall		<u>Free fall from all sides:</u> Height of fall: 0.4 meter onto steel. Actuator not activated/connected.
Vibration	EN 60068-2-36 (Fdb) EN 60068-2-6 (Fc)	<u>Random vibration:</u> Short time test: 6.29 g RMS Actuator is not connected. Long time test: 7.21 g RMS Actuator is not connected. Duration: 2 hours in each direction <u>Sinus vibration:</u> Frequency 5-25 Hz: Amplitude = 3.3 mm pp Frequency 25-200 Hz: Acceleration 4 g Number of directions: 3 (X-Z-Y) Duration: 2 hours in each direction. Actuator is not activated.
Bump	EN 60068-2-29 (Eb)	<u>Bump test:</u> Level: 40 g Duration: 6 milliseconds Number of bumps: 500 shocks in each of 6 directions. Actuator is not connected.
Shock	EN 60068-2-27 (Ea)	<u>Shock test:</u> Level: 100 g Duration: 6 milliseconds Number of bumps: 3 shocks in each of 6 directions. Actuator is not connected.

Environmental tests - Electrical

Test	Specification	Comment
Emission	EN 61000-6-4	Level is within limits for the motor
Automotive Transients	ISO 7637	Load dump test only accepted on motor power connection.
IECEX / ATEX (Ex)	EN 60079-0:2012 EN 60079-31:2014	This Ex certification allows the actuator to be mounted in Ex dust areas: II 2D Ex tb IIIC T135 °C Db Tamb -25 °C to +65 °C

Environmental tests – Climatic - LA36 Off-highway

Test	Specification	Comment
Cold Test	EN 60068-2-1 (Ab)	Storage at low temperature: Temperature: -55 °C Duration: 72 h Actuator is not activated/connected Tested at room temperature.
	EN 60068-2-1 (Ad)	Operating at low temperature: Temperature: -40 °C Duration: 16 h Actuator is not activated/connected. Tested at low temperature.
Dry Heat	EN 60068-2-2 (Bb)	Storage at high temperature: Temperature: +105 °C Duration: 72 h Actuator is not activated/connected. Tested at room temperature
	EN 60068-2-2 (Bd)	Storage at high temperature: Temperature: +70 °C Duration: 1000 h Actuator is not activated/connected Tested at high temperature. Operating at high temperature: Temperature: +60 °C Int. max. 17 % Duration: 700 h Actuator is activated Tested at high temperature.
Change of Temperature	EN 60068-2-14 (Na)	Rapid change of temperature: High temperature: +105 °C in 60 minutes. Low temperature: -40 °C in 60 minutes. Transition time: <10 seconds Duration: 117 cycles Actuator is not activated/connected. Tested at room temperature.
	EN 60068-2-14 (Nb)	Controlled change of temperature: Temperature change 5 °C pr. minute High temperature: +70 °C in 60 minutes. Low temperature: -30 °C in 30 minutes. 130 minutes pr. Cycle. Duration: 1.000 cycles (90 days) Actuator is not activated/connected. Tested at 250, 500 and 1,000 cycles at low and high temperatures.

Test	Specification	Comment
Damp Heat	EN 60068-2-30 (Db)	Damp heat, Cyclic: Relative humidity: 93-98 % High temperature: +55 °C in 12 hours Low temperature: +25 °C in 12 hours Duration: 21 cycles * 24 hours Actuator is not activated/connected. Tested within 1 hour after condensation, after upper temperature has been reached.
	EN 60068-2-3 (Ca)	Damp heat, Steady state: Relative humidity: 93-95 % Temperature: +40 ± 2 °C Duration: 56 days Actuator is not activated/connected. Tested within one hour after exposure.
	EN 600068-2-78	Temp 40 °C Relative Humidity 95% Test Duration 168 Hours Actuator is not activated/connected. DUT orientation: Normal operating orientation
Salt Mist	ISO 9227	Continuously salt spray test: Salt solution: 5 % sodium chloride (NaCl) Actuator activated/connected. Exposure time: 500 hours
Degrees of Protection	EN 60529 – IP54	IP5X - Dust: Dust-tight, No ingress of dust. Actuator is not connected IPX4 – Water: Ingress of water in quantities causing harmful effects is not allowed. Duration: 100 litres pr. minute in 3 minutes Actuator is not connected
	EN 60529 – IP66	IPX6 –Connected actuator: Actuator is driving out and in for 3 min. 100 (l/min) jet of water is placed at the wiper ring for 3 (min). IPX6 –Connected actuator and push 6800 (N) Actuator is preheated to 85 °C driving out and in for 3 min. and Push 6800 (N) at the end-pos. 100 (l/min.) jet of water (15 °C) is placed at the wiper ring for 3 min.
	ISO 20653 – IP6KX	Category : 1 (air pressure reduction) Dust medium : Talcum Air pressure : 2 kPa (20 mbar) below normal air pressure Duration : 8 hours.
	ISO 20653 – IPX9K	High pressure cleaner: Water temperature: +90 °C Water pressure: 350 bar Flowrate: 15L/min Spray angle: 360 ° Spray distance: 200 mm Duration: From any direction continuously in 300 seconds Actuator is connected. Ingress of water in quantities causing harmful effects is not allowed.

Test	Specification	Comment		
	ISO 16750	The actuator has been warmed up to 85 °C for 8 hours. After this it is cooled down in 00 °C salt water. Cooling time: 120 minutes Repeated 5 times and weight recorded after each cycle. Opened for check of residue of salt deposit and water. No weightgain, water or salt residue allowed.		
Chemical test	Chemicals:	Explanation regarding chemical (poss. incl. source of supply and/or trade name):	Test temp °C:	Test duration:
	Diesel fuel	EN 590	85	22 h
	'Bio' diesel fuel	EN 14214	85	22 h
	Engine oil	Multigrade oil SAE OW40, API SL / C	85	22 h
	Transmission fluid	ATF Dexron III'	85	22 h
	Hydraulic fluid	DIN 51 524-3 (HVLP ISO VG 46)	85	22 h
	Grease	DIN 51 502 (KP2K-30)	85	22 h
	Anti-freeze 50 % (1:1)	Ethylene glycol (C2H6O2) - Water mixture	85	22 h
	Urea Nitrogen saturated solution	ISO 22241-1; Urea NOx "ad blue" e.g. P3 Solvclean AK (manufactured by Henkel)	85	22 h
	Cold cleaner	E.g. P3 Solvclean AK (by Henkel)	RT	22 h
	Contact spary	E.g. WD 40	85	22 h
	Ammonium hydroxide	20% diluted water-based solution	RT	22 h
	Liquid lime 10 % (Super- Cal)	Lime fertiliser, 10% solution; can be obtained from seed and fertiliser traders	RT	22 h
	NPK Fertilizer (NPK 16-4-12) saturated	Nitrogen (5-9%), phosphorus (5-9%) potassium (5-9%) portions can be obtained from seed and fertiliser traders e.g. COMPO 14361/14354, PhytoGreen NPK 8-8-6 or similar.	RT	22 h

Environmental tests - Mechanical - LA36 Off-highway

Test	Specification	Comment
Free Fall	EN 60068-2-31	Free fall from all sides: Height of fall: 0.45 meter onto concrete Actuator not activated/connected.
Vibration	EN 60068-2-64	Random vibration: Test: 5.9 g RMS 10-2,000 Hz Actuator is connected for monitoring during the test Duration: 32 hours in each direction (x,y and z).
Shock	EN 60068-2-27 (Ea)	Shock test: Level: 51 g Duration: 6 milliseconds Number of bumps: 10 shocks in each of 6 directions. Actuator is connected for monitoring during the test

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